EC 571: Digital VLSI Circuit Design

Introduction

Ajay Joshi

Acknowledgement - Some slides adapted from:

- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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What will you learn in EC571?

- Digital integrated circuit design
  - Understanding, designing and optimizing digital circuits and large VLSI systems that meet different specifications: speed, power, cost and reliability

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Detailed topics

- CMOS manufacturing technology
- CMOS devices, gates
- Design parameters: delay, noise margins, area, power
- Combinational and sequential circuits
- Interconnect
- Timing and clock
- Memories
- Arithmetic building blocks
- CAD tools / Design methodologies

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Logistics

- Class schedule
  - Lecture: Monday and Wednesday – 6:30 pm to 8:15 pm in PHO 205
  - Recitation: Friday – 2:30 pm to 4:15 pm in PHO 305
- Class website
  - Blackboard – http://learn.bu.edu
    - I'll post course material and other useful links on Blackboard
  - Piazza – http://www.piazza.com
    - We'll use Piazza for announcement and questions
  - Gradescope – http://www.gradescope.com
    - We'll use Gradescope for submitting and grading homeworks
- Print only what is necessary
  - http://www.bu.edu/sustainability/

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Course textbooks

- Textbooks
  - Textbook (strongly recommended): **Digital Integrated Circuits – A Design Perspective**, Second Edition, by J. Rabaey, A. Chandrakasan, B. Nikolic
  - TextBook (optional): **Digital VLSI Chip Design with Cadence and Synopsys CAD Tools**, First Edition. Erik Brunvand
  - TextBook (optional): **CMOS VLSI Design: A Circuit and Systems Perspective**, Fourth Edition, Neil Weste and David Harris
- Research papers
  - ACM digital library: http://www.acm.org/dl/
  - IEEE Explore: http://ieeexplore.ieee.org/Xplore/home.jsp

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Teaching staff information

- Instructor
  - Prof. Ajay Joshi
    - Email: joshi@bu.edu
    - Office hours: By appointment in PHO 334
- Graduate Teaching Assistant
  - Yang Wu
    - Email: yangwu@bu.edu
    - Office hours: Tuesday and Thursday 8 pm ET to 9 pm ET in PHO 305
- Grader/Lab Assistant
  - Thinh Phuc Nguyen
    - Email: ptnguy01@bu.edu

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Grading policy

- Three mid-term exams (20% each)
  - October 14, 2026
  - November 11, 2026
  - December 9, 2026
- Project (16%)
  - Details will be discussed in class
- Homework (24%)
- Class participation will help your grade

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Academic Misconduct

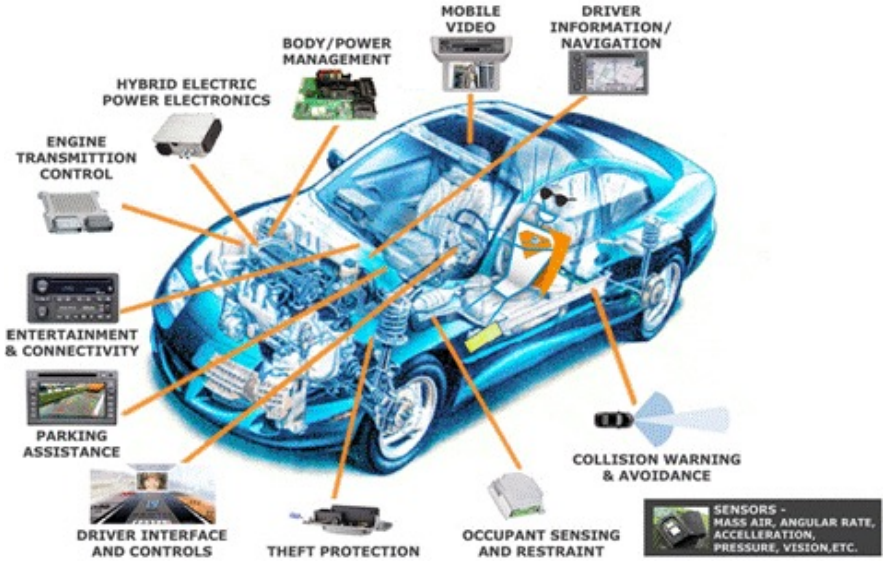
- Cheating will **not** be tolerated
- General rule:
  - Anything with your name on it must be **YOUR OWN** work
    - Example: Individual work on homework assignments
  - Reference any source you use in your work
    - Example: textbook, research paper, website
- Possible penalties
  - Zero on assignment (minimum)
  - Fail course
  - Note on permanent record
  - Suspension
  - Expulsion
- BU Honor code applies

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Electronic chips are everywhere!

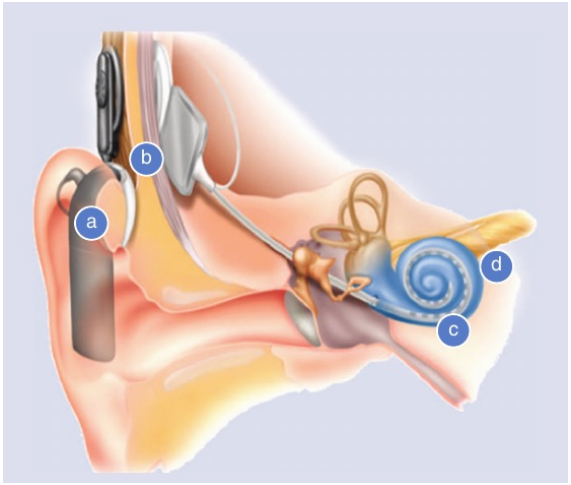


www.wcspittsburg.org



www10.edacafe.com

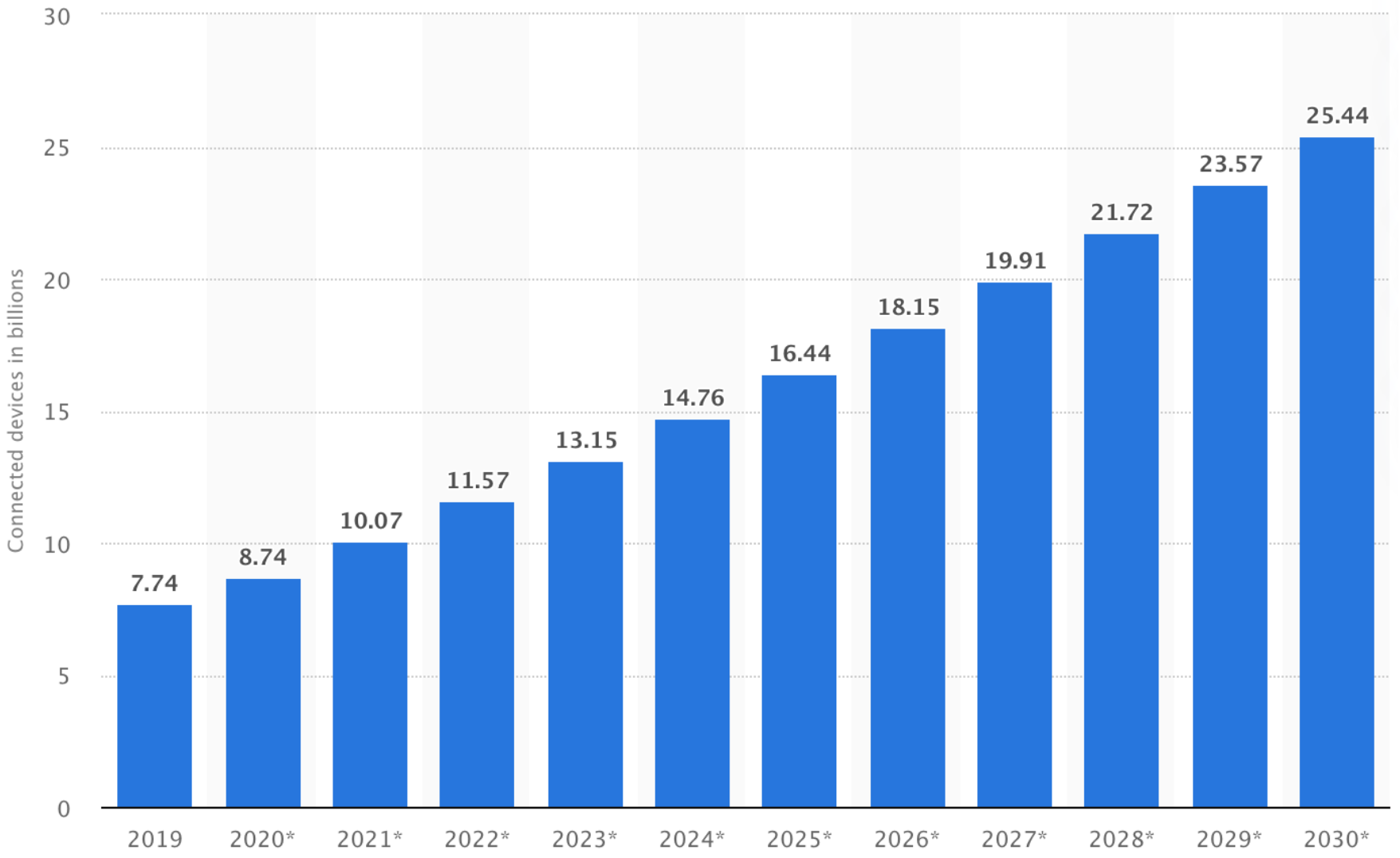
lifesciences.ieee.org



www.extremetech.com



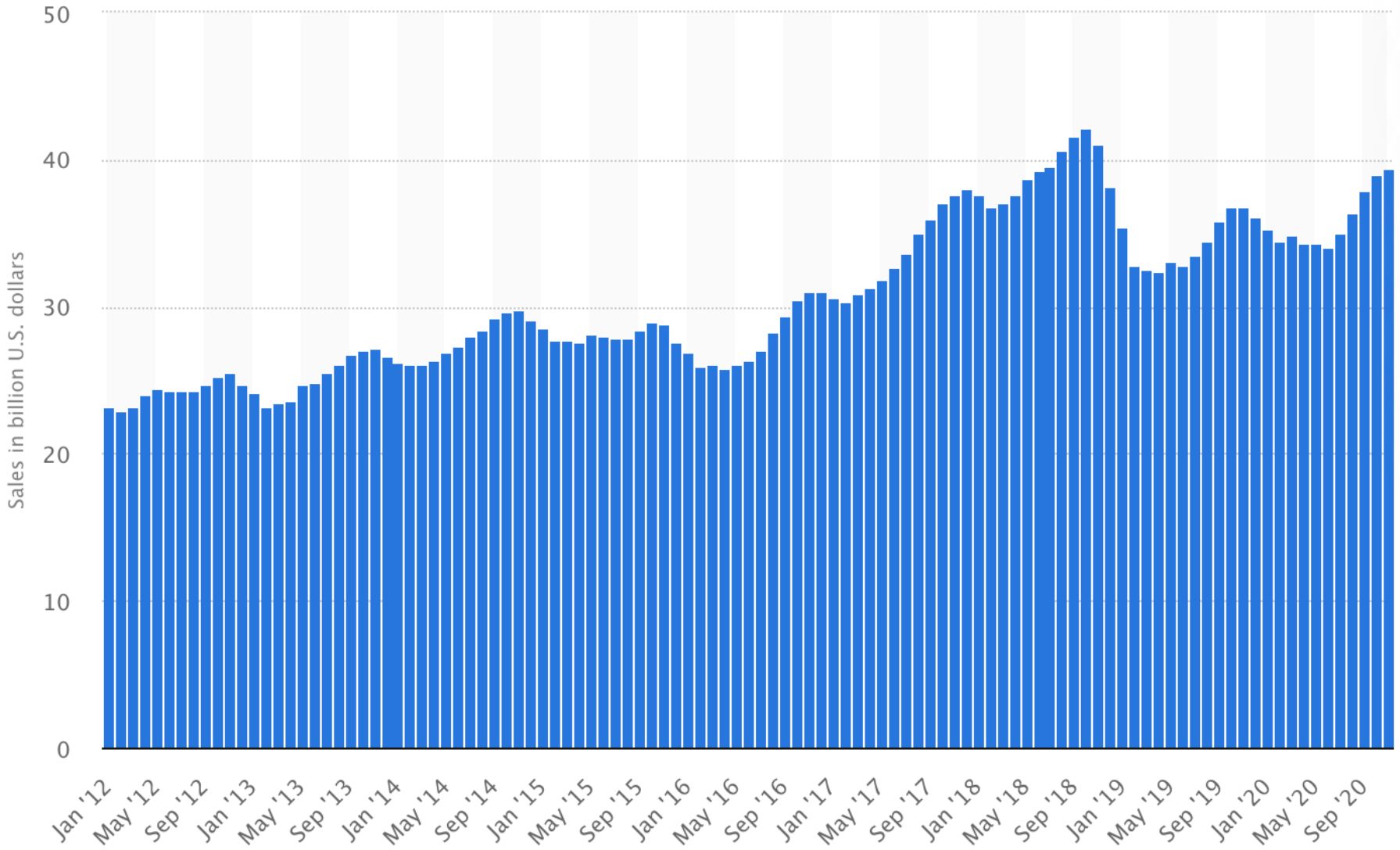
IoT devices will dominate the market



Source: Statista

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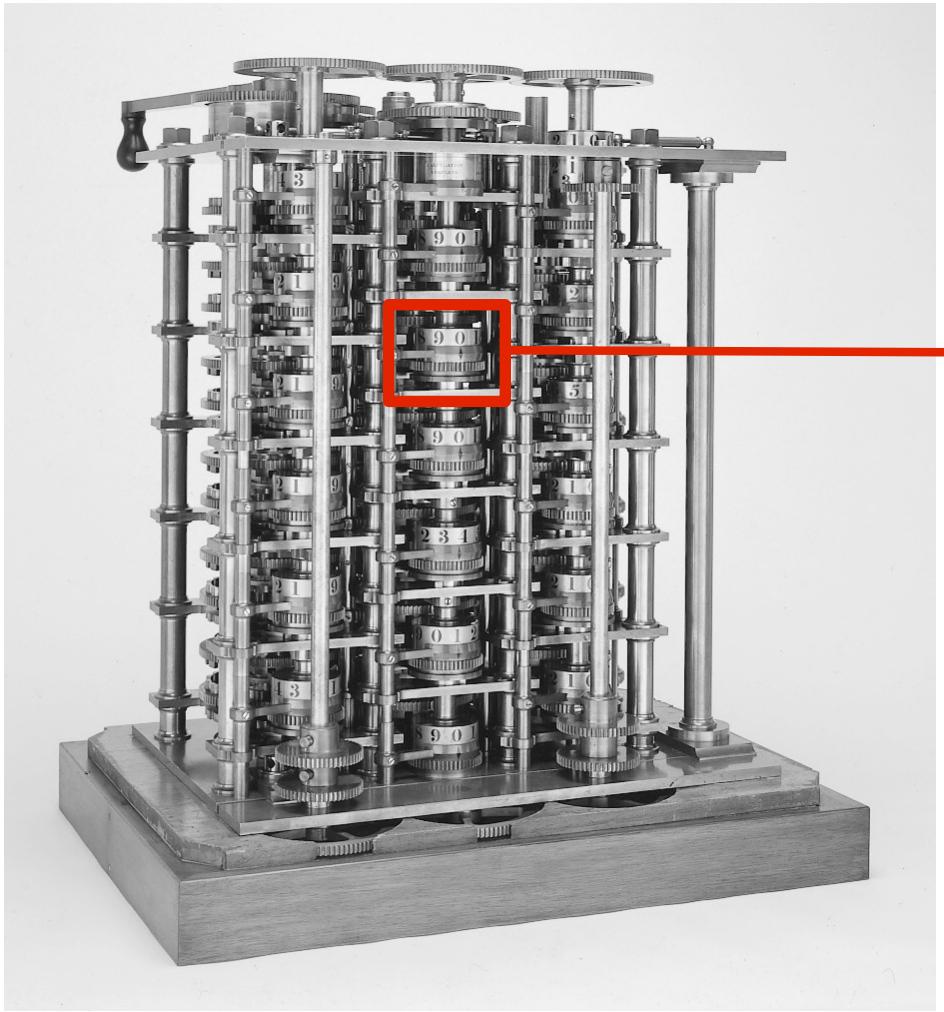
Semiconductor Sales Worldwide



Source: Statista

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The First Computer





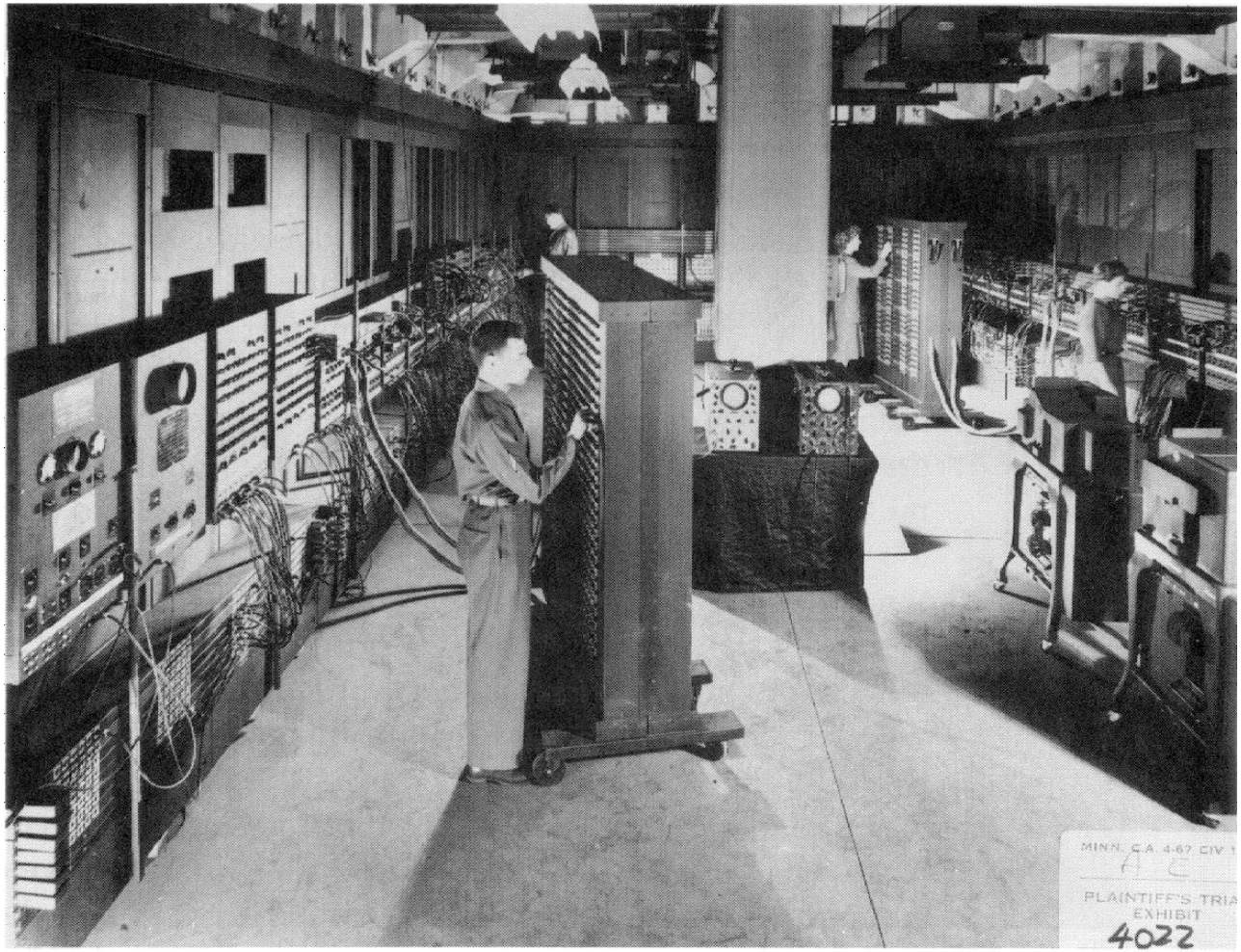
The Babbage Difference Engine (1832)

25,000 parts

cost: £17,470

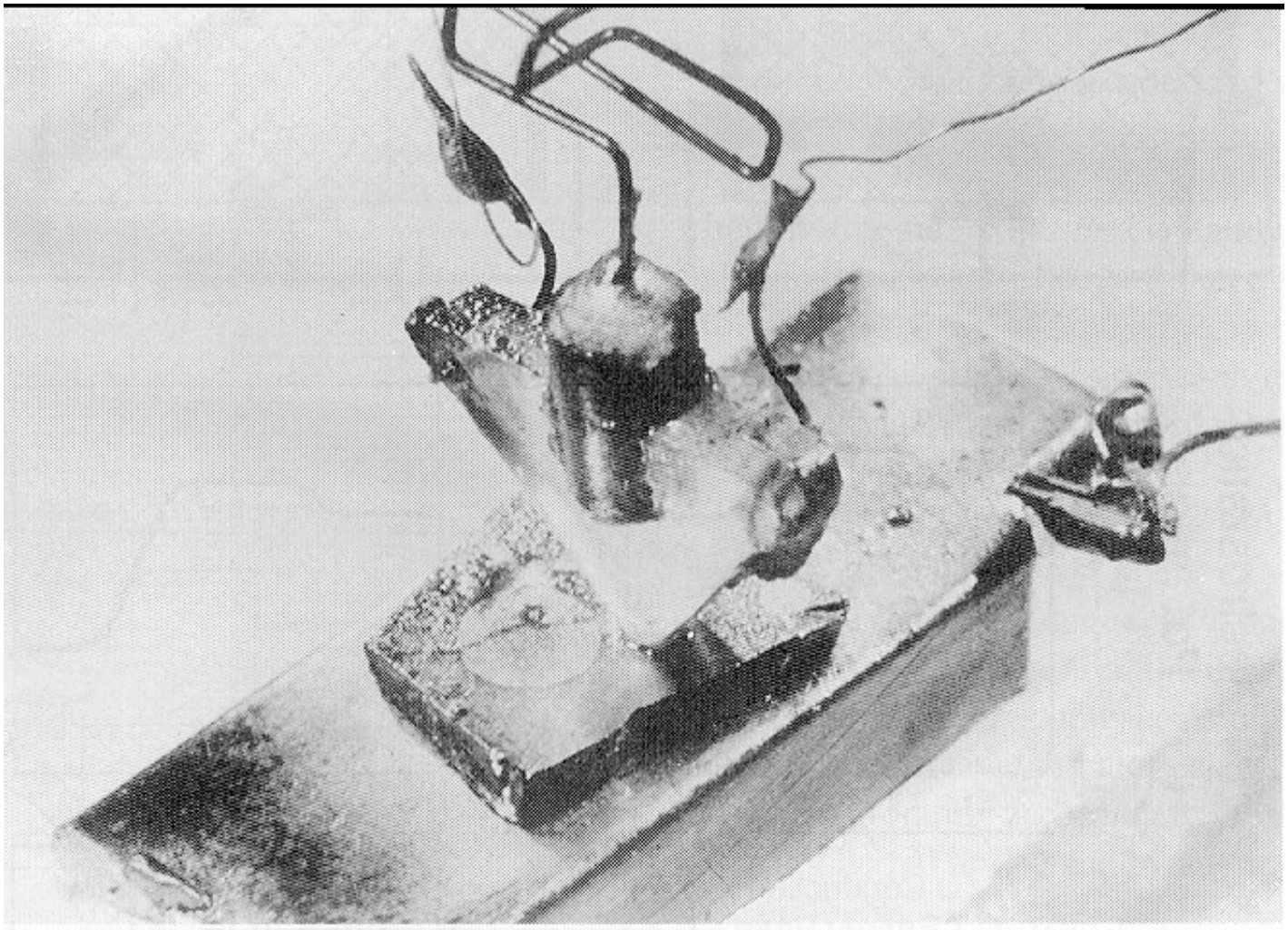
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ENIAC - The first electronic computer (1946)



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The Transistor Revolution

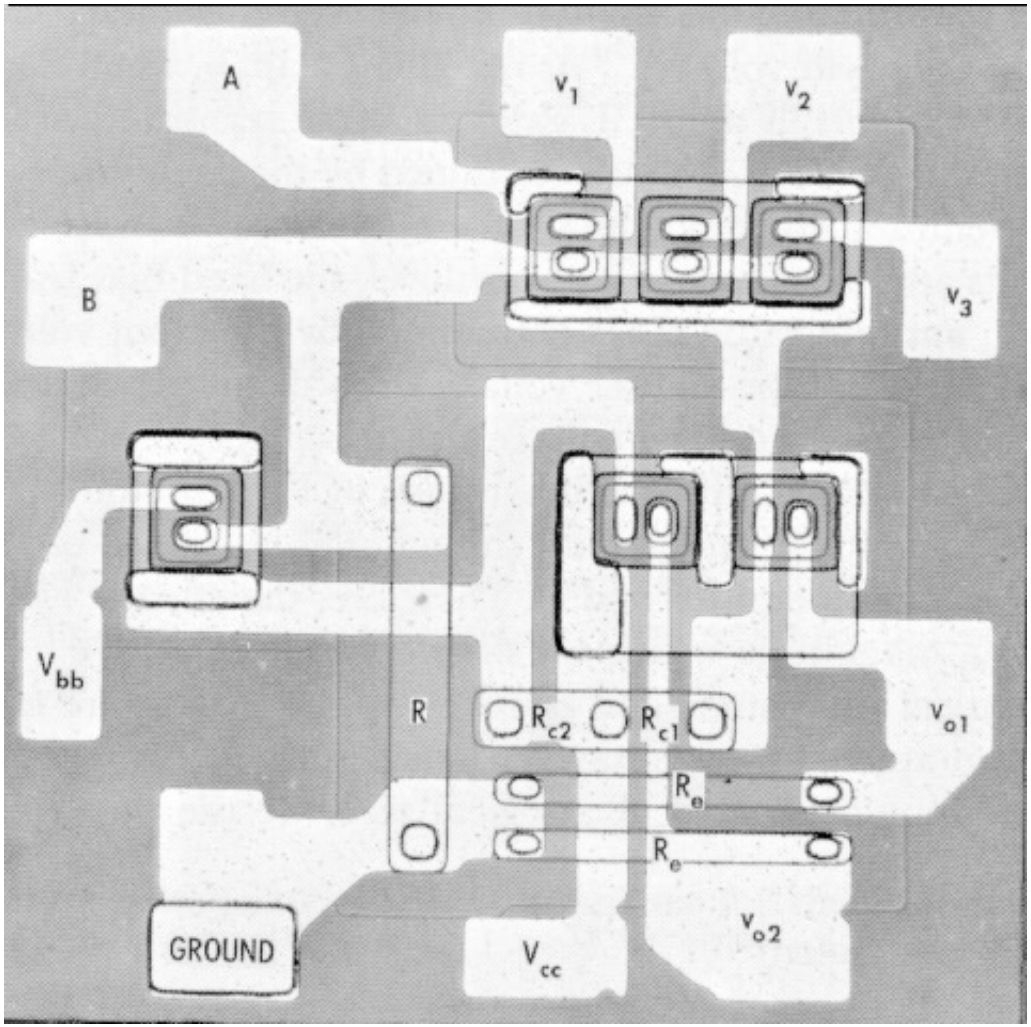


First transistor
Bell Labs, 1948

First transistor invented by Bardeen, Shockley and Brattain

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The First Integrated Circuits



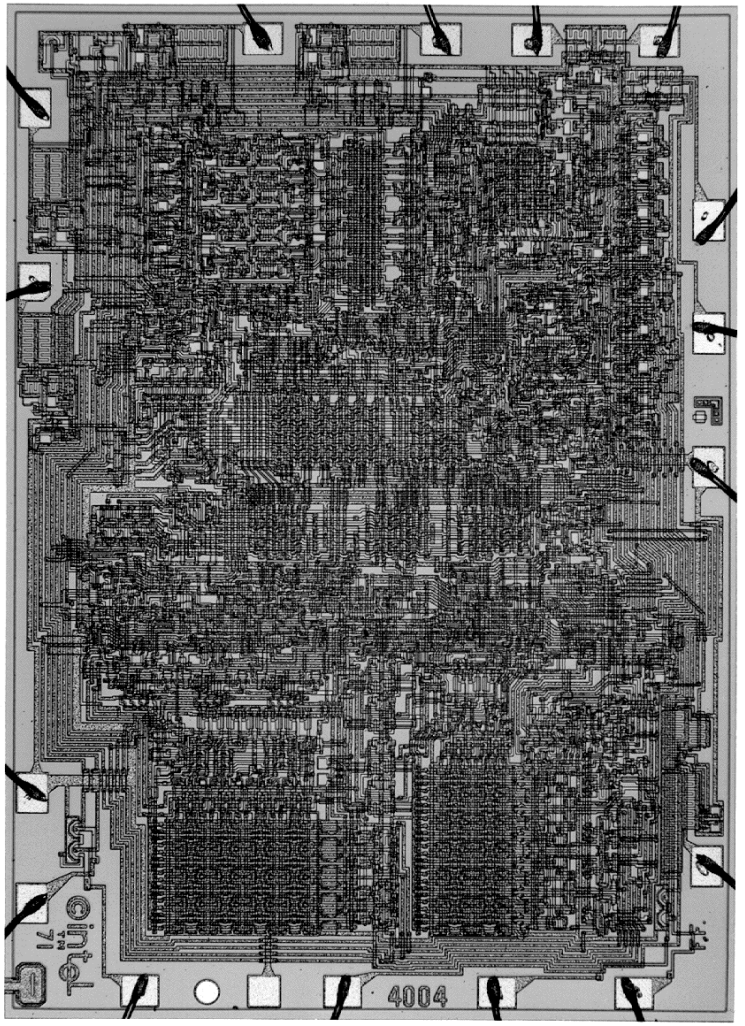
Bipolar logic
1960's

ECL 3-input NAND Gate
Motorola 1966

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Intel 4004 Micro-Processor (1971)

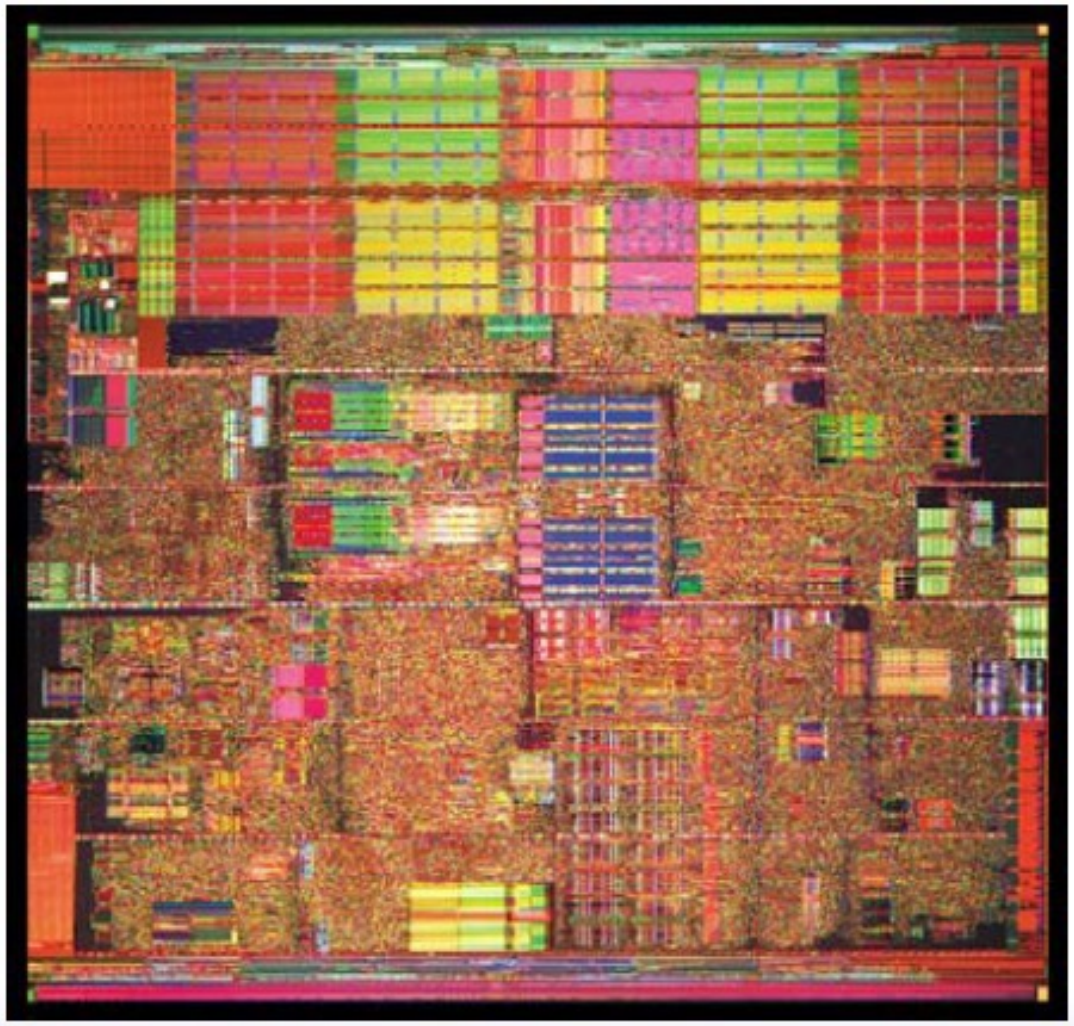
- 2,300 transistors (12 mm^2)
- 740 KHz operation
- 10 µm PMOS technology



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Intel Pentium-4 microprocessor (2000)

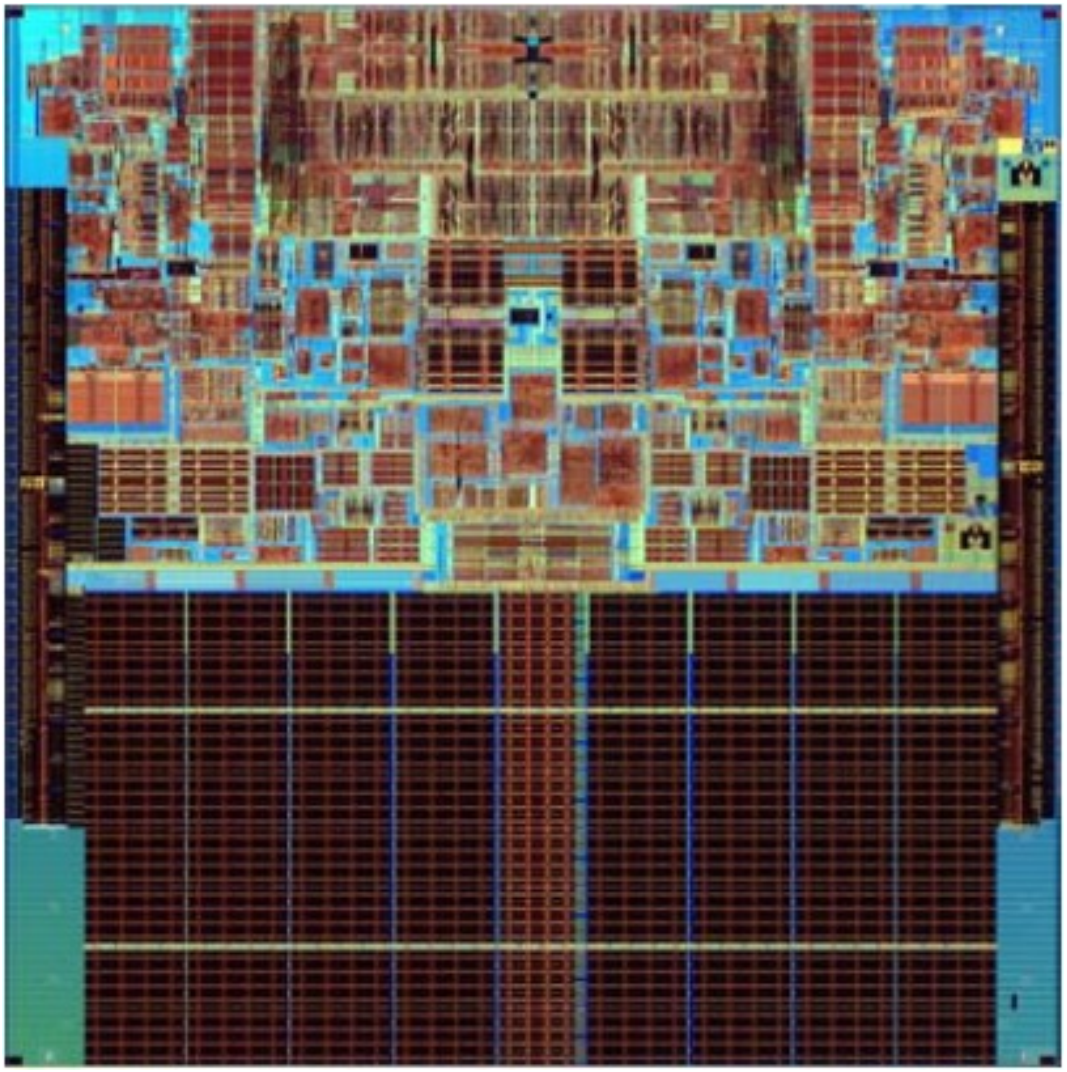
- 125,000,000 transistors (112 mm^2)
- 3.8 GHz operation
- 90 nm CMOS technology



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Intel Core 2 microprocessor (2006)

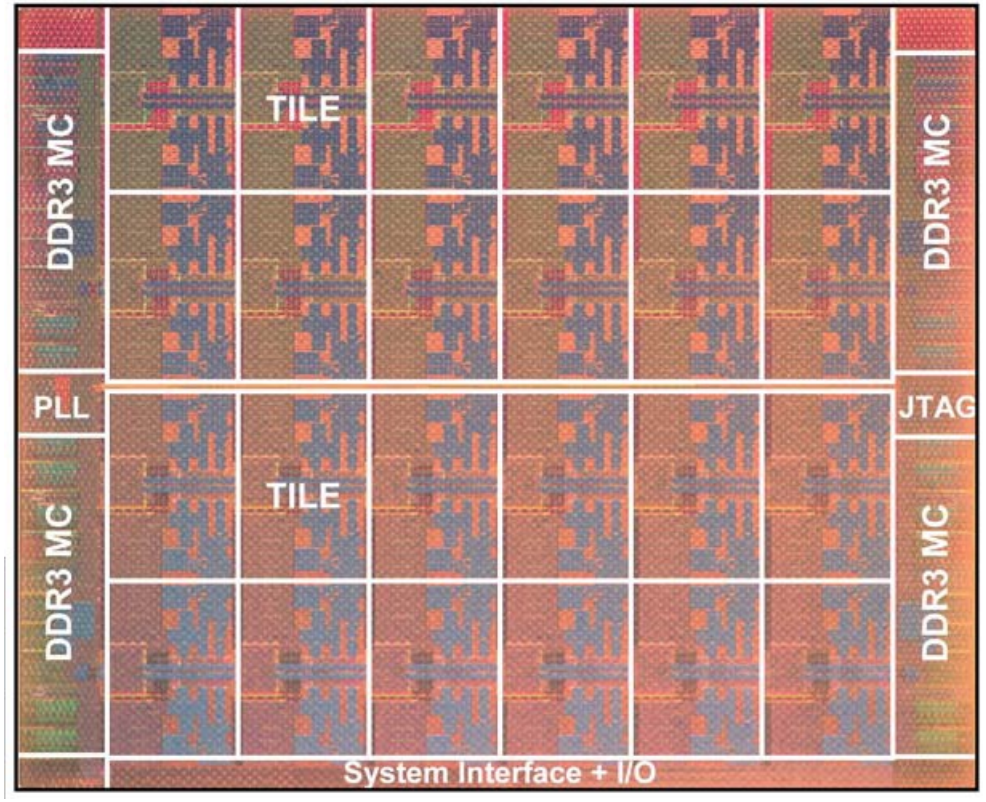
- 291,000,000 transistors (143 mm^2)
- 3 GHz operation
- 65 nm CMOS technology



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Intel Single Chip Cloud (SCC) processor (2010)

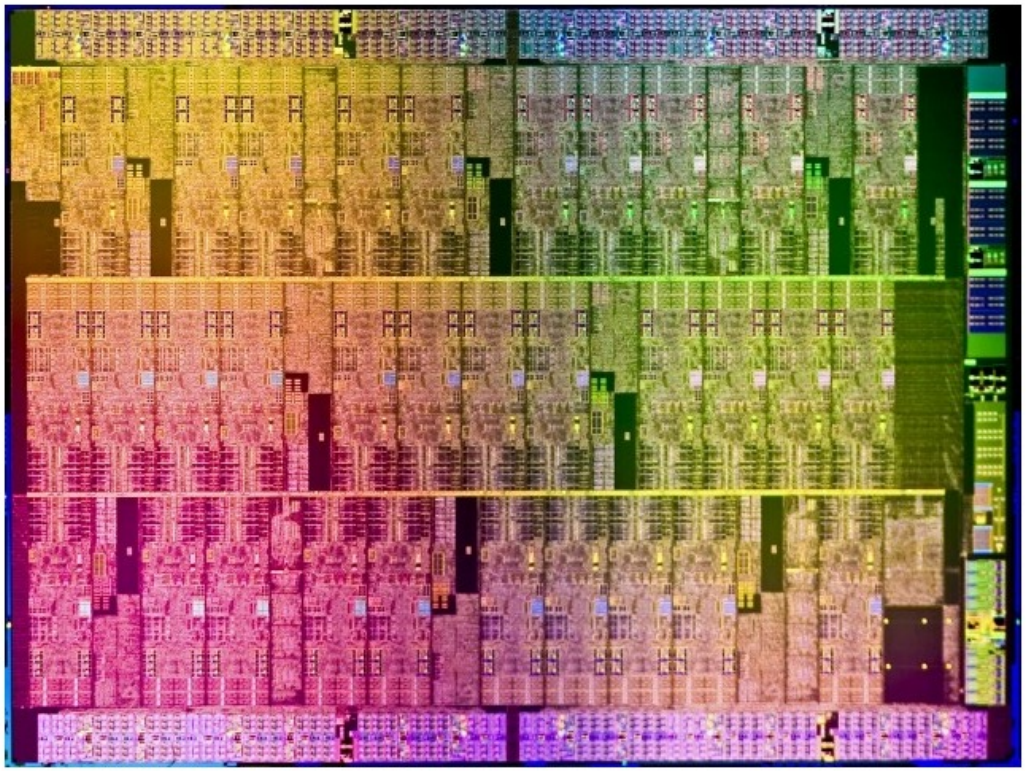
- 1,300,000,000 transistors (567 mm^2)
- 48 cores
- 3 GHz operation
- 45 nm CMOS technology



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Intel Xeon Phi processor (2015)

- ~8,000,000,000 transistors (400-500 mm^2)
- 72 cores
- 1-1.5 GHz operation
- 14 nm CMOS technology



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Intel Xeon Platinum 9282 processor (2019)

- ~8,000,000,000 transistors
- 56 cores
- 2.6 GHz operation
- 14 nm CMOS technology

?

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Intel i9-13900KS (2023)

- ~14,200,000,000 transistors
- 24 cores
- Up to 6 GHz operation
- 7 nm CMOS technology

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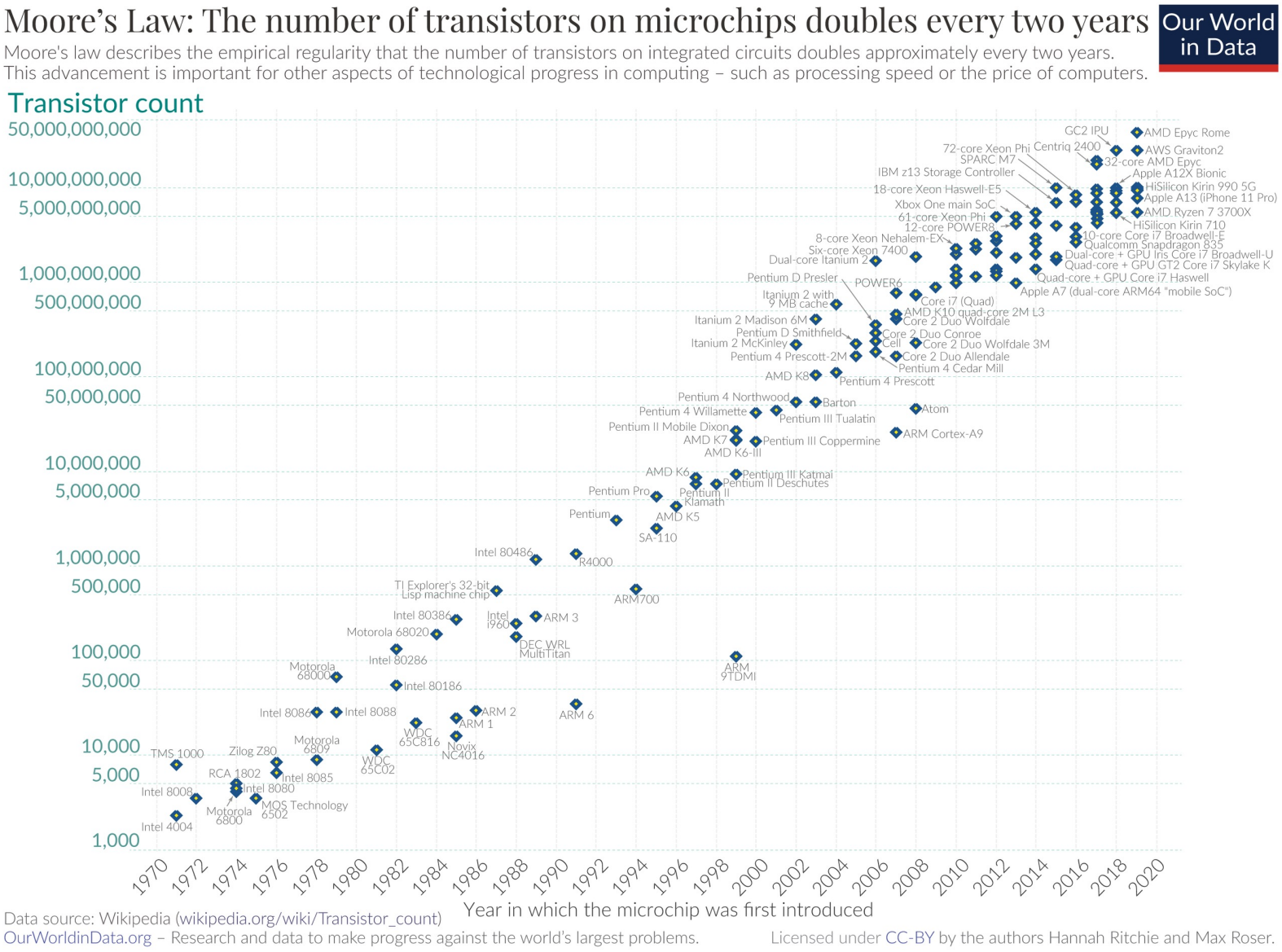
Moore's law

- In 1965, Gordon Moore noted that the number of transistors on a chip doubled every 18 to 24 months.
- He made a prediction that semiconductor technology will double its effectiveness every 18 months

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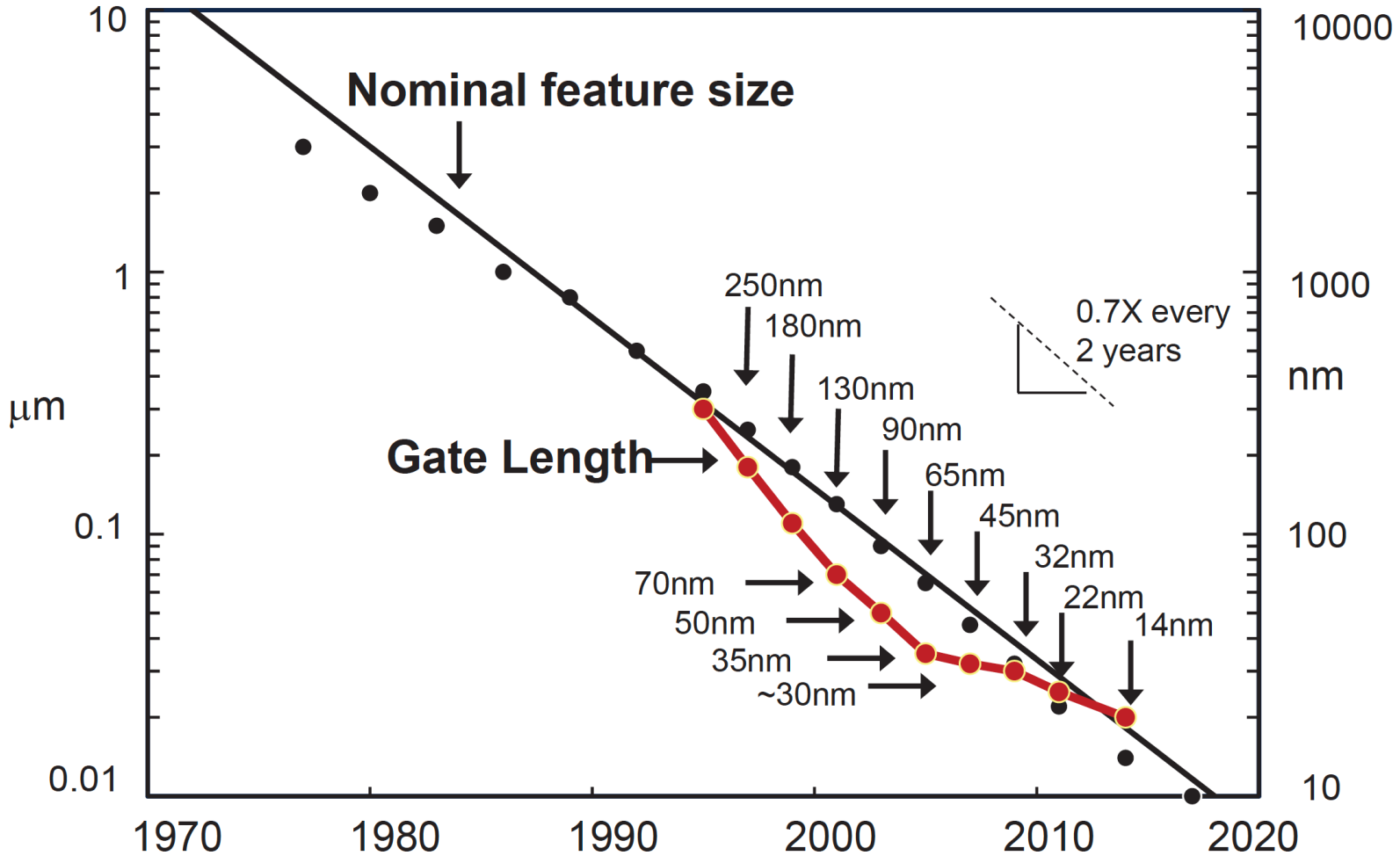
Moore’s law - Transistor Count

Source: https://en.wikipedia.org/wiki/Transistor_count



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Printed vs. Physical Gate

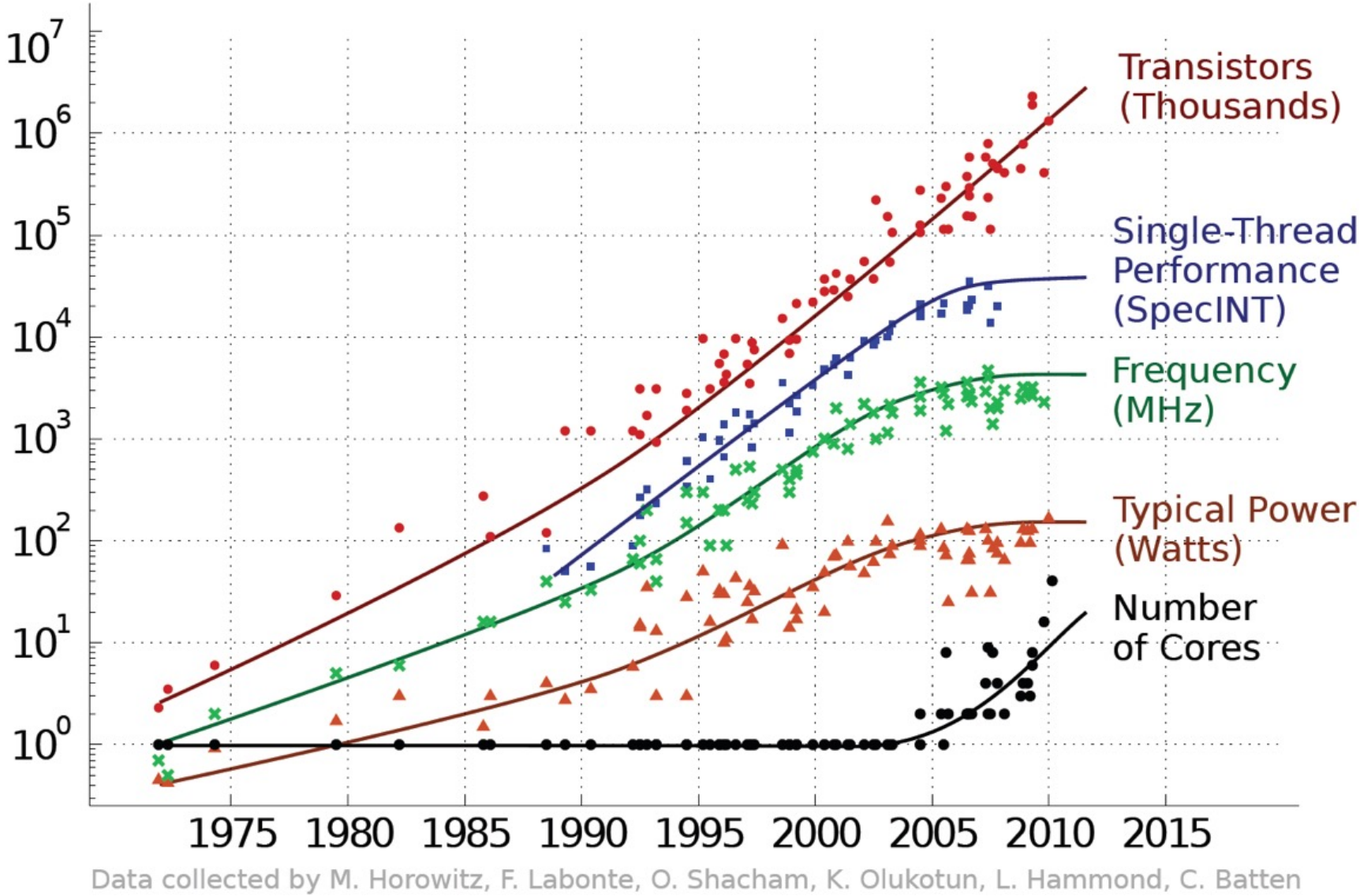


Physical gate length > nominal feature size after 22nm

Source: Intel, IEDM presentations

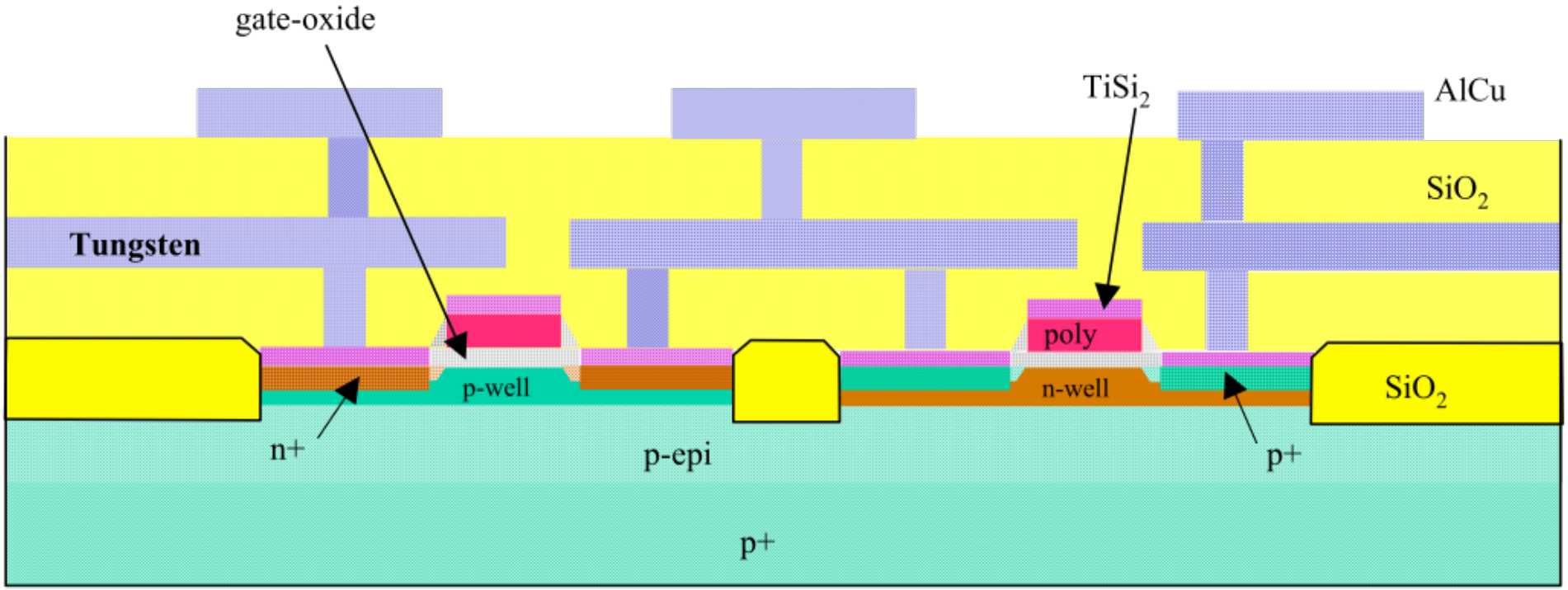
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Trends in VLSI systems



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A Typical CMOS Process



Dual-Well Trench-Isolated CMOS Process

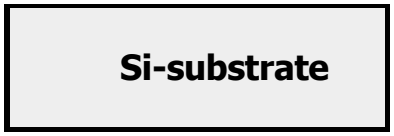
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Basic Fabrication Steps

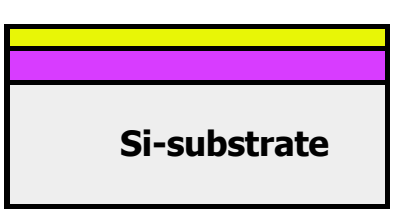
- Transfer an image of the layer design to the wafer
- Do something to the imaged part of the wafer
  - Implant, Deposit, Grow, Etch, etc
- Strip off image and sometimes polish wafer to make it flat
- Key points
  - Entire die image transferred at once
  - Entire wafer surface processed at once
  - Economical because everything done in parallel

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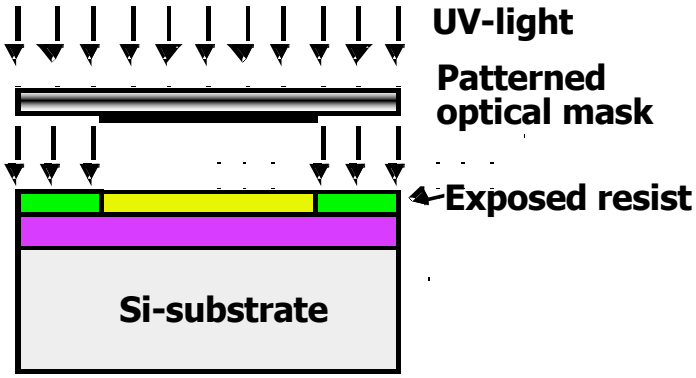
Patterning of SiO_2



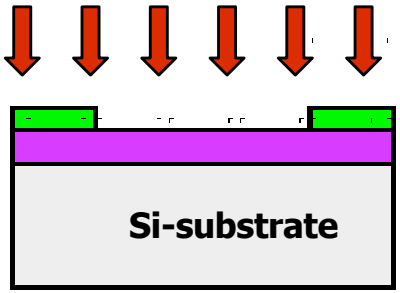
(a) Silicon base material



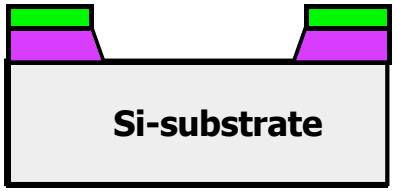
(b) After oxidation and deposition of negative photoresist



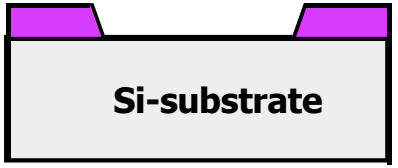
(c) Stepper exposure



(d) After development and etching of resist, chemical or plasma etch of SiO_2



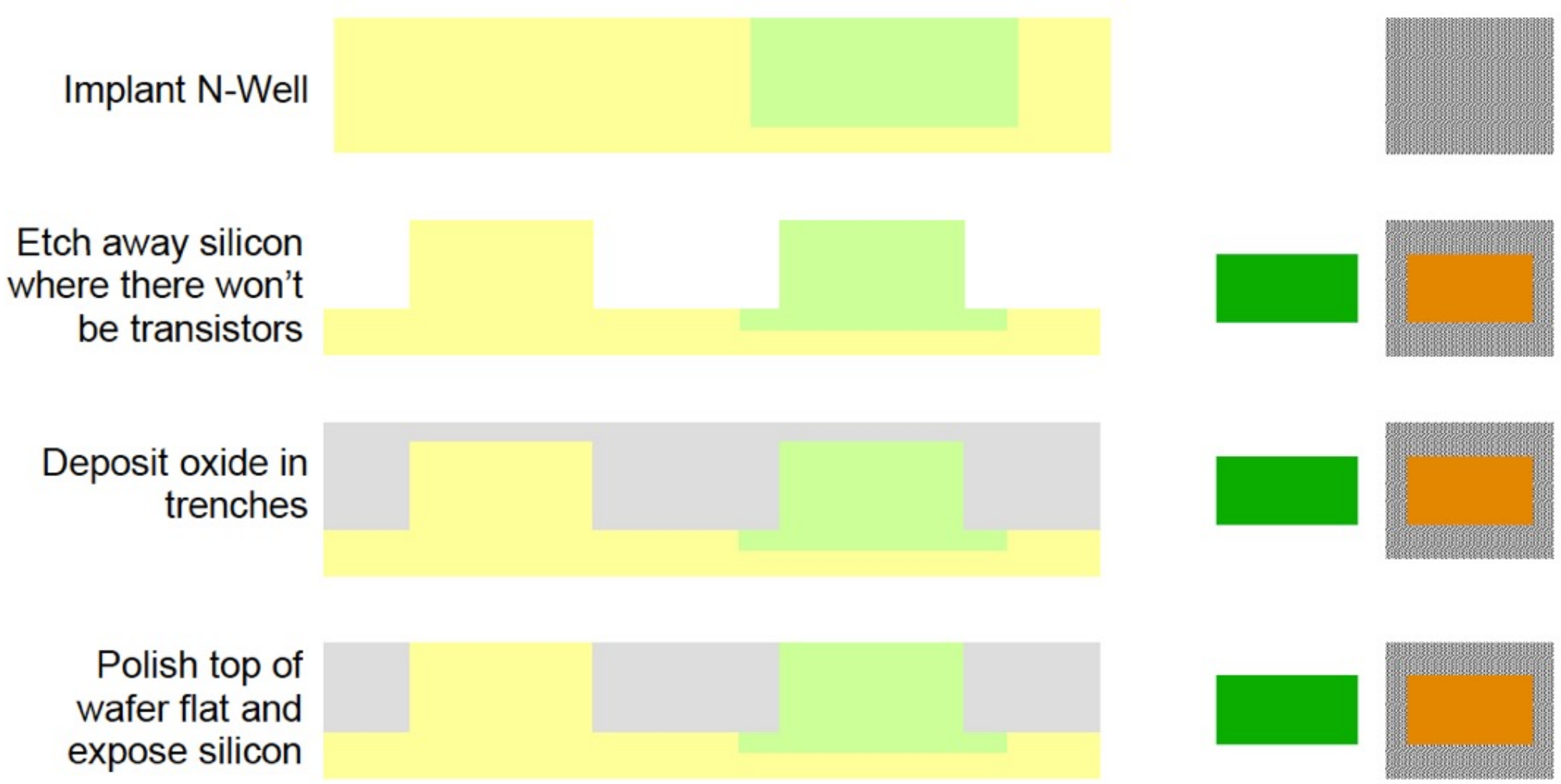
(e) After etching



(f) Final result after removal of resist

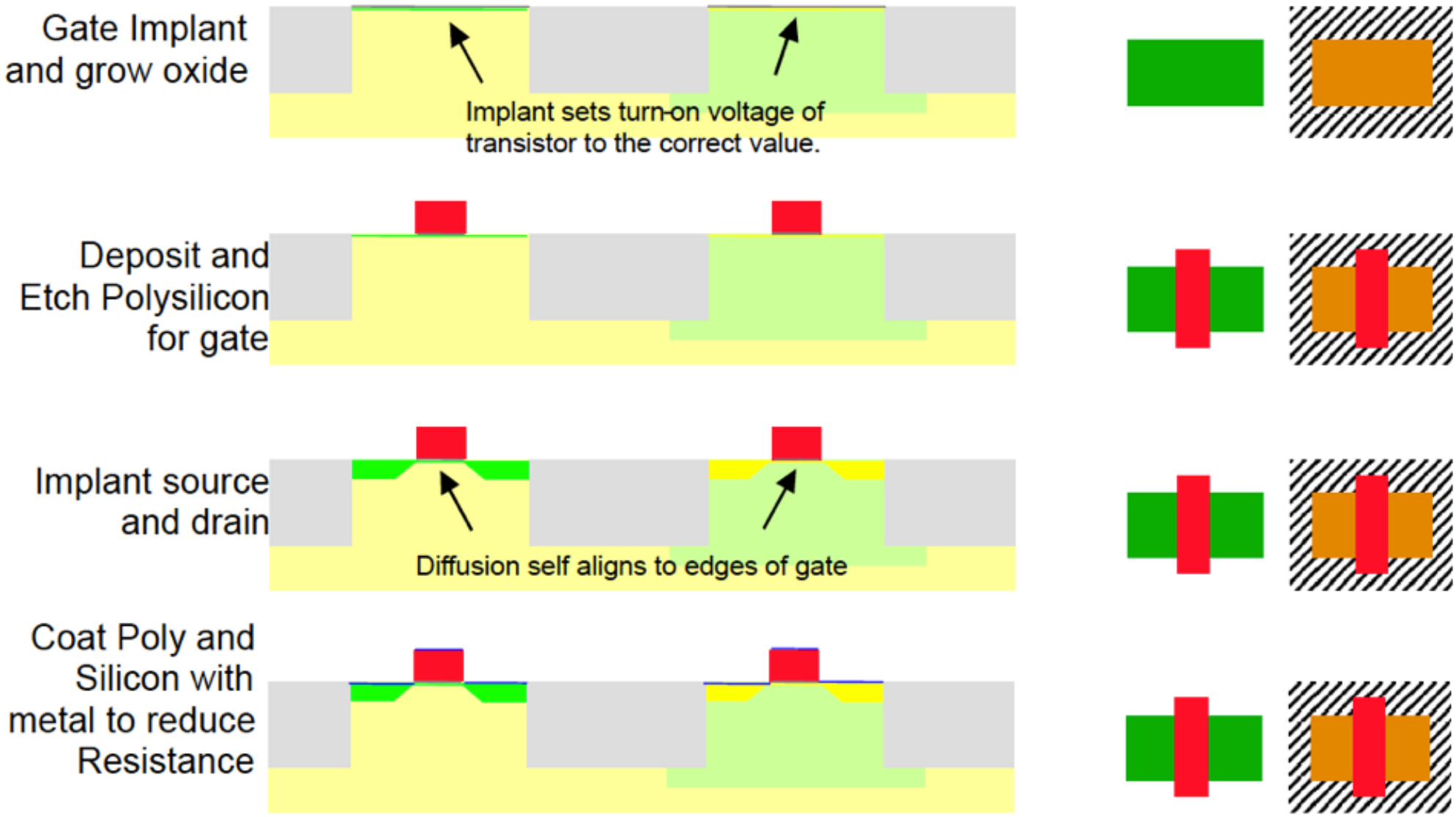
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Making a transistor



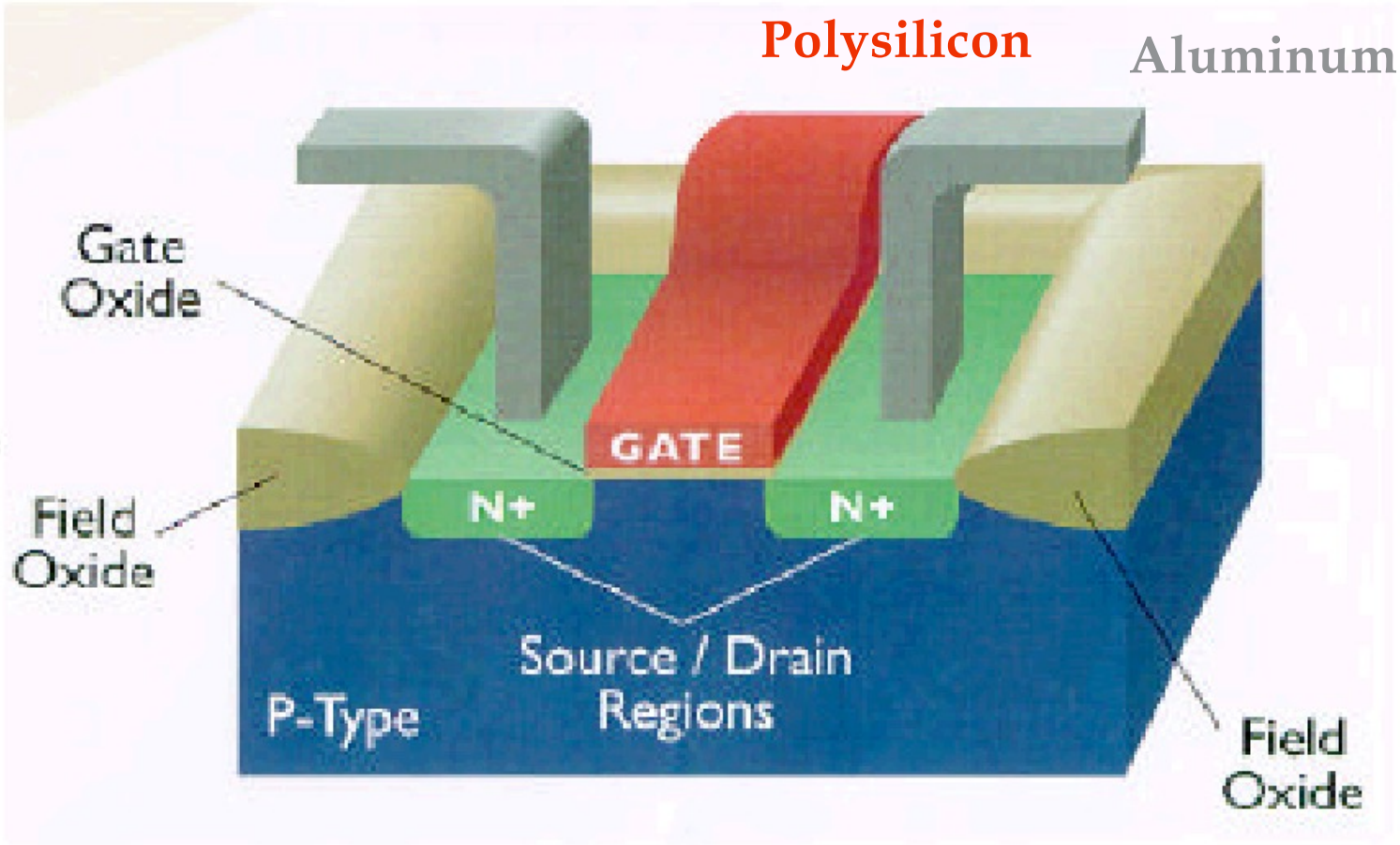
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Making a transistor



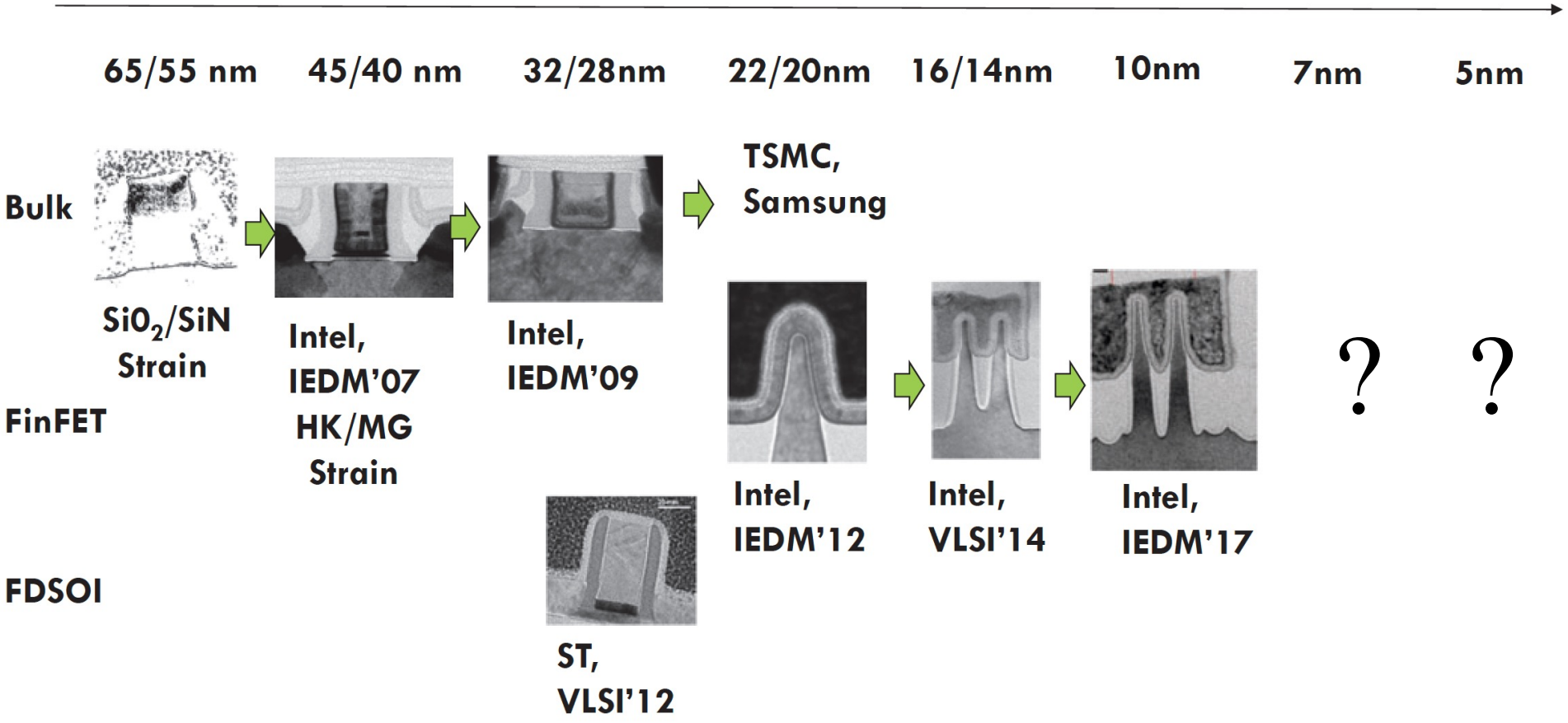
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3D Perspective



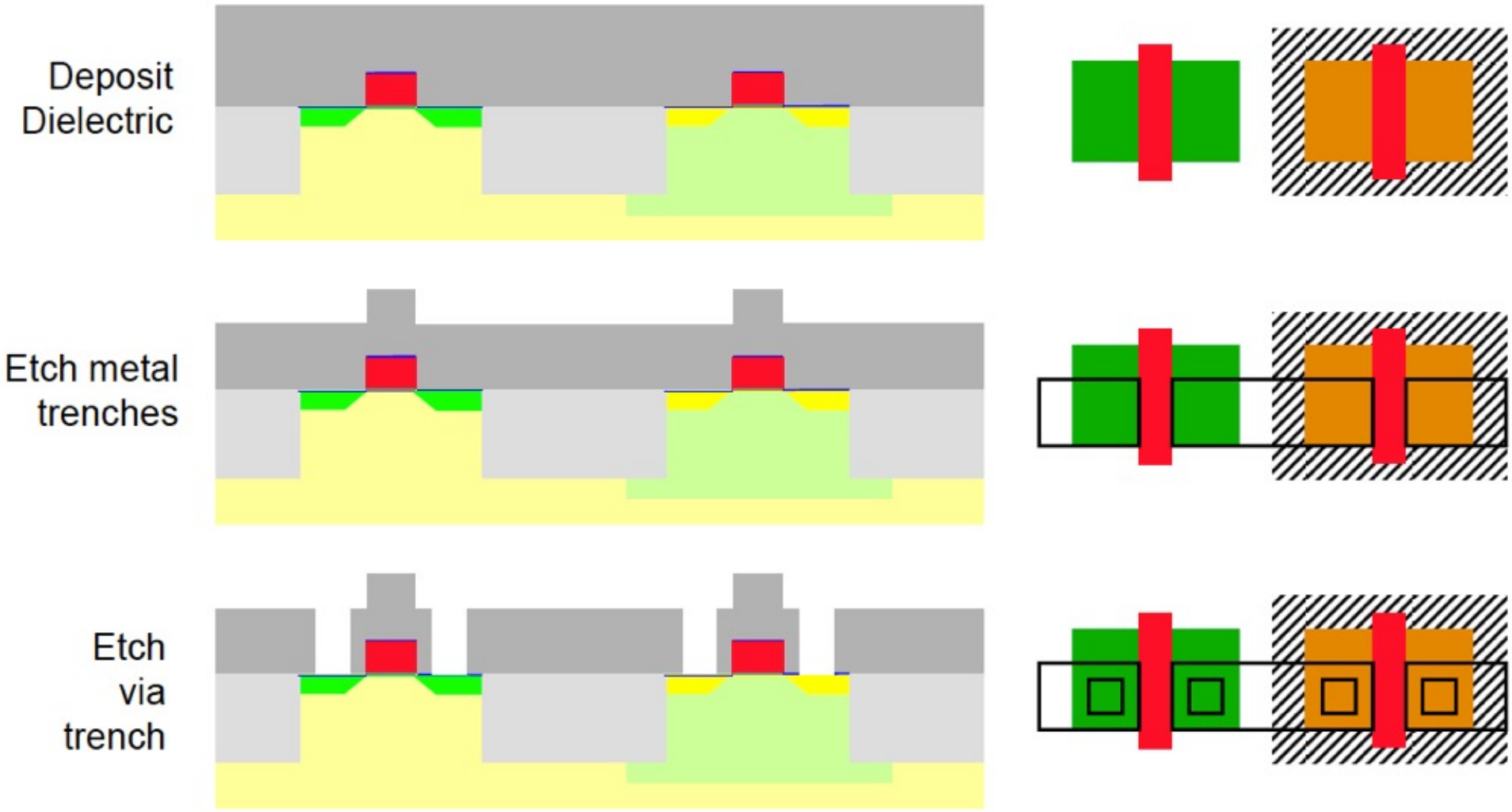
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Transistors are Changing



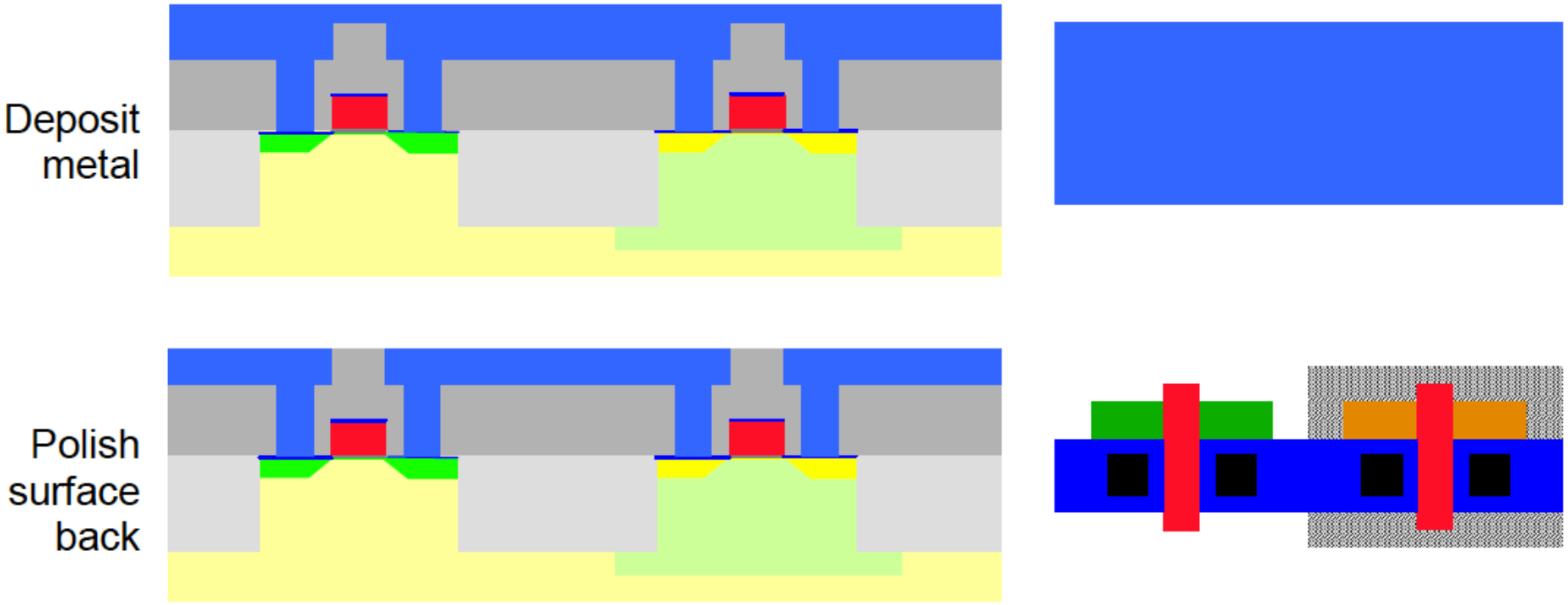
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Making a wire



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Making a wire



Same process can be repeated to add additional metal layers

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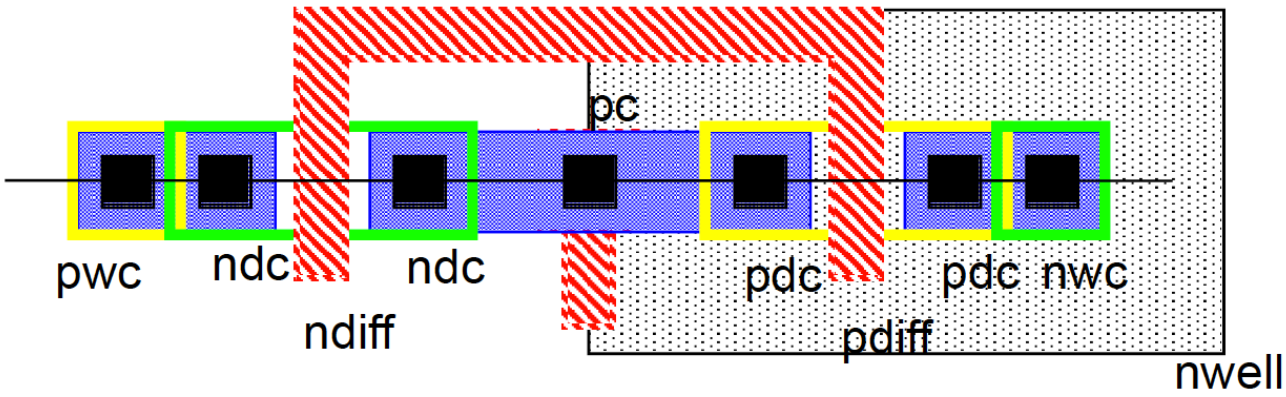
Intel Fabrication Process

- http://www.youtube.com/watch?v=d9SWNLZvA8g
- https://www.youtube.com/watch?v=_VMYPLXnd7E

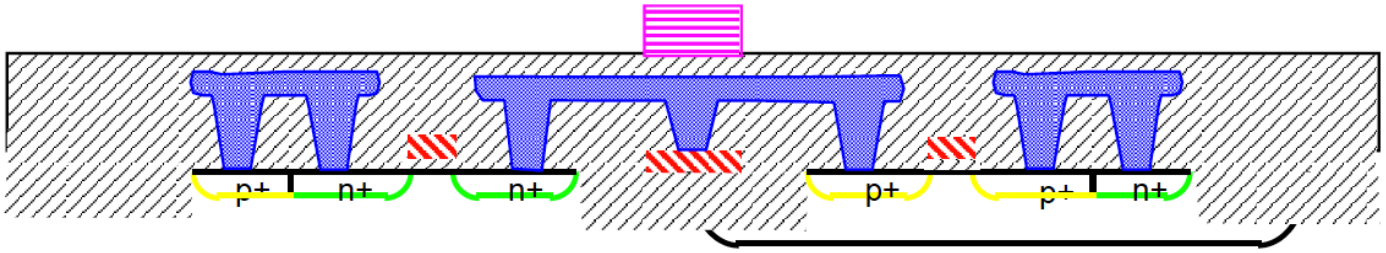
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What do we do?

- Draw artwork like this



- So fab can create something like this (cross-sectional view)



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Fabrication Layout Constraints

- Fundamental constraints
  - Resolution
    - Smallest width feature
    - Smallest spacing without shorts
  - Alignment/Overlap
    - Need to align layers with each other
    - Tolerance required to be manufacturable
- Other constraints
  - Density rules
  - …

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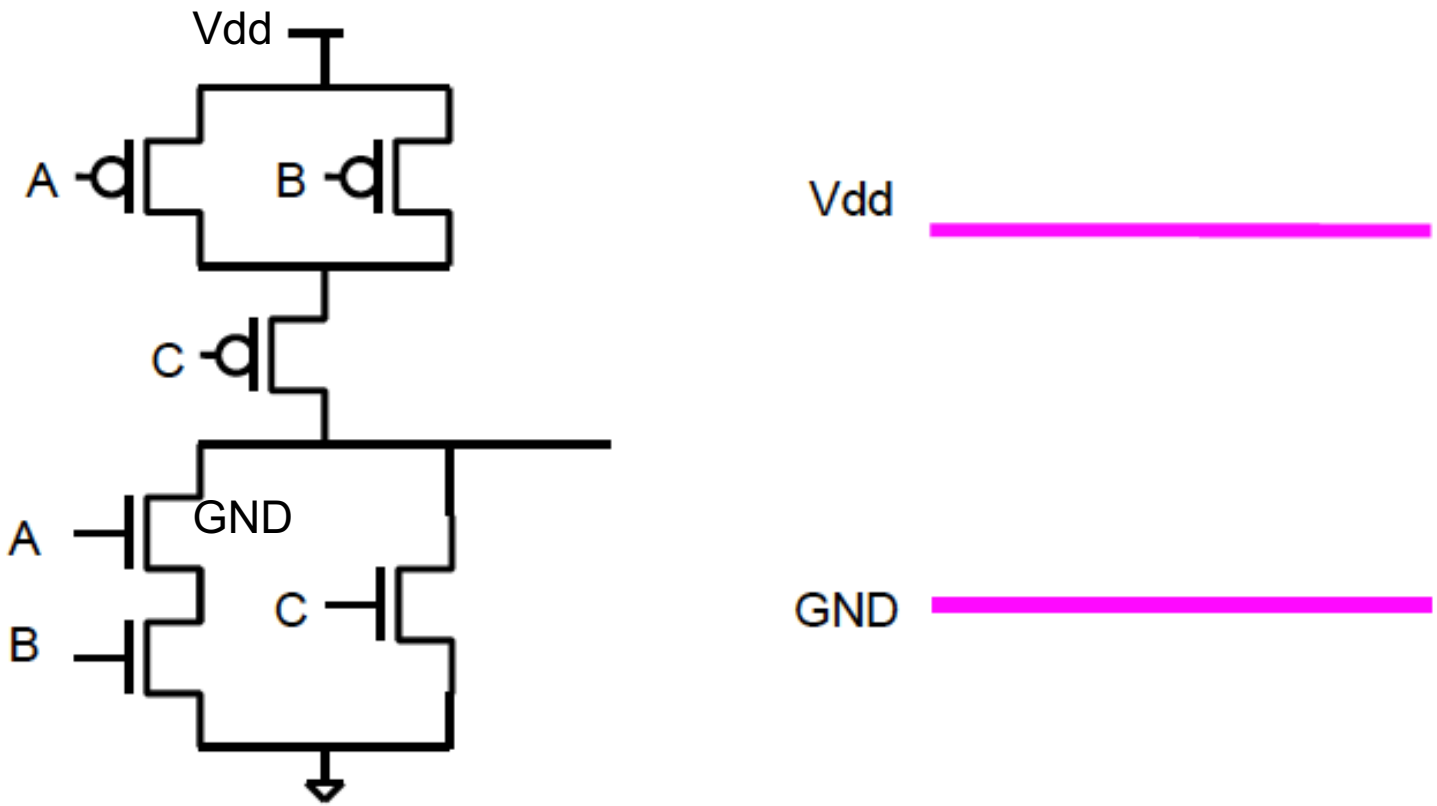
Design Rules

- Layout constraints are formulated to form design rules
- Interface between designer and process engineer
- Guidelines for constructing process masks
- Example Design Rules
  - Minimum poly width/spacing
  - Minimum active region width/spacing
  - Minimum metal width/spacing
  - Contact region size/spacing

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Example - Step 1

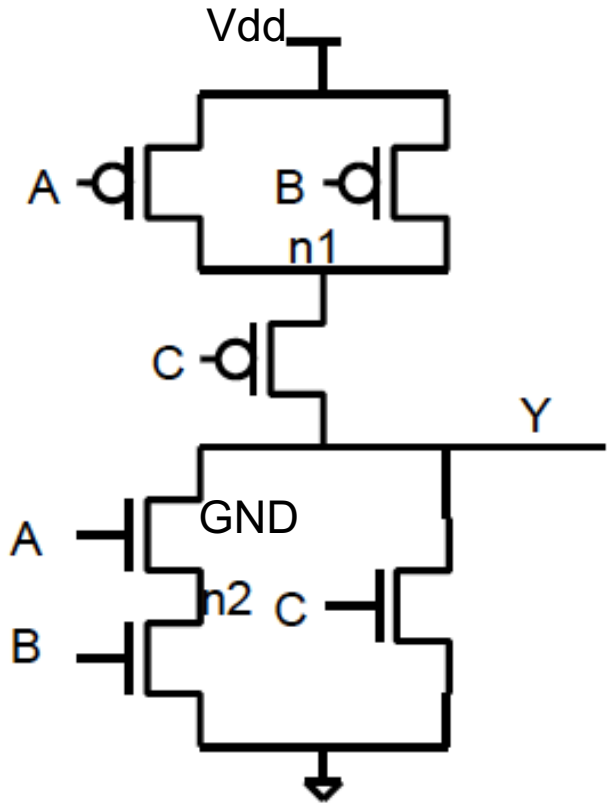
- Choose global directions for routing layers
- Position power lines in top layer of metal

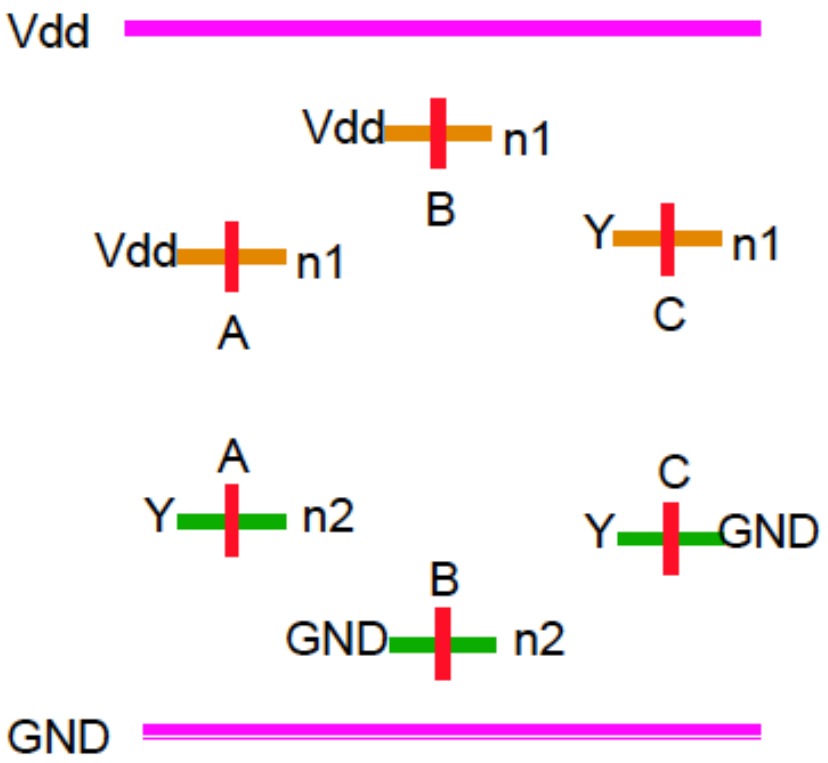


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Example - Step 2

- Cluster together NMOS with NMOS and PMOS with PMOS
- Keep gate orientation the same

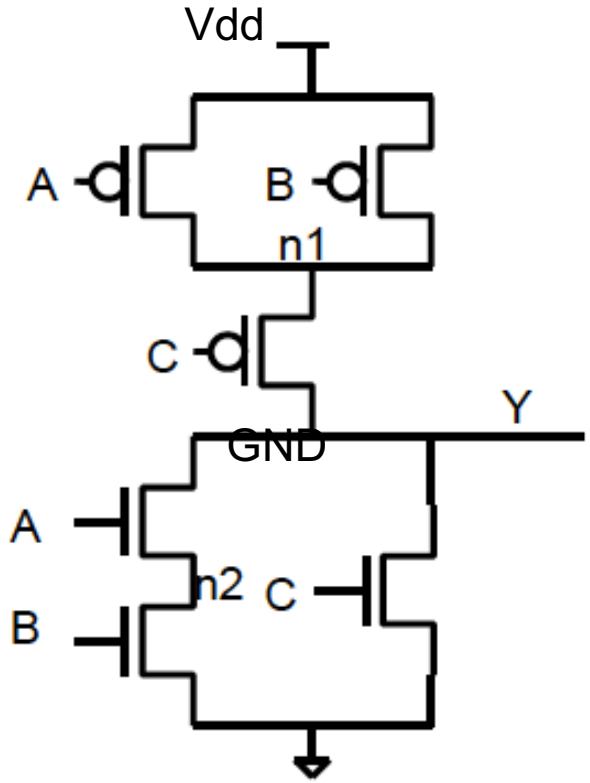


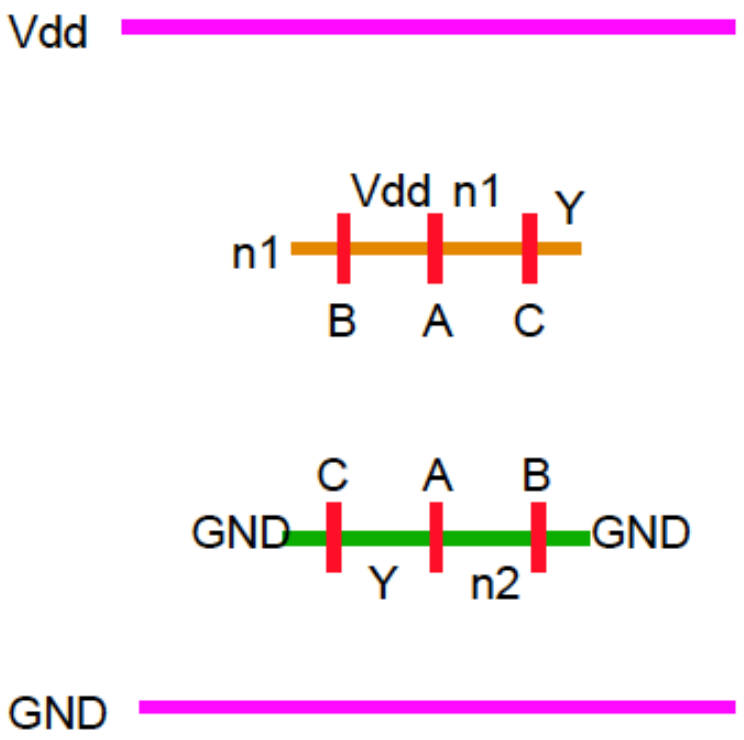


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Example - Step 3

- Arrange transistors so that common sources/drains can be shared
- Give precedence to shared signals over shared vdd/ground

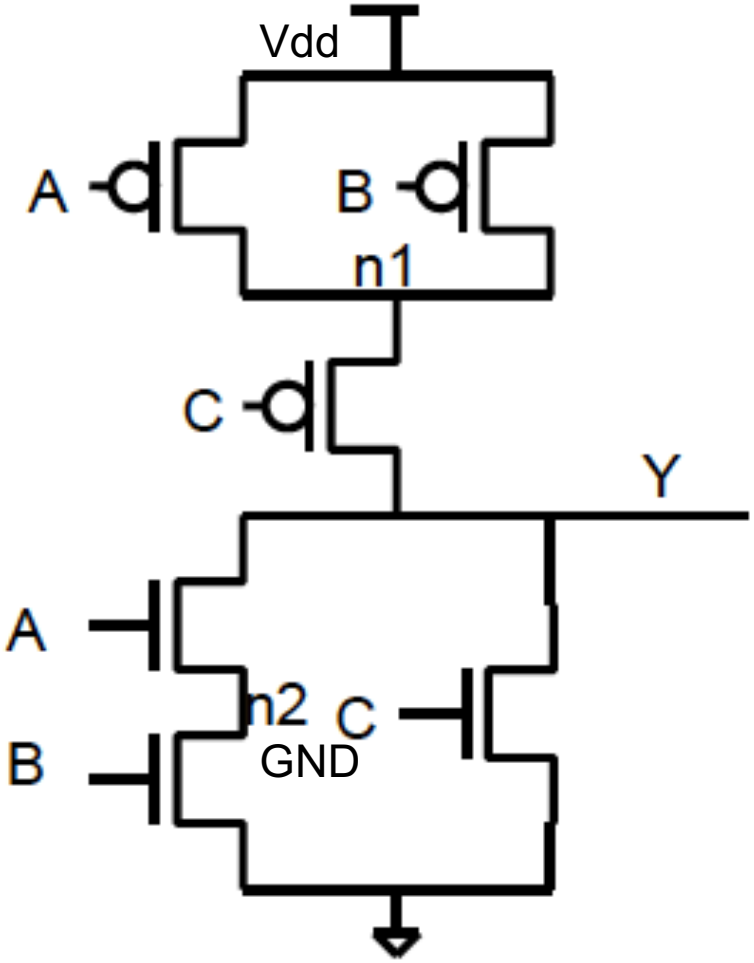


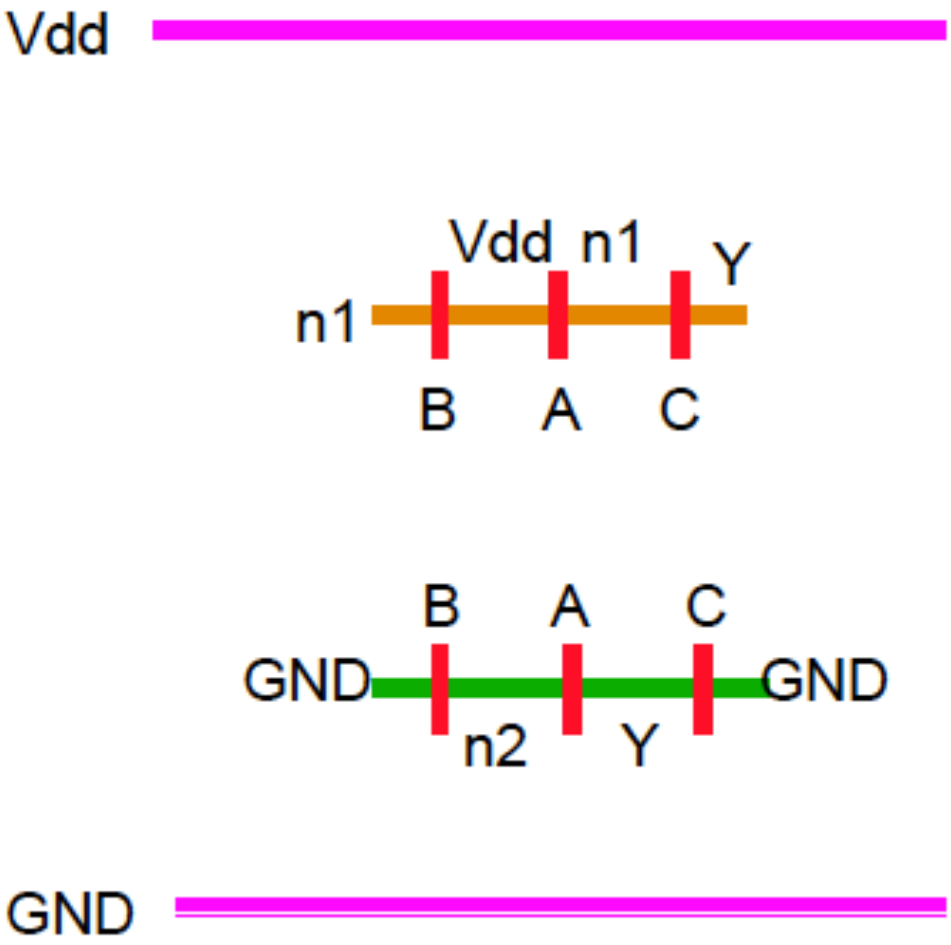


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Example - Step 4

- Arrange transistors so that common gates line up

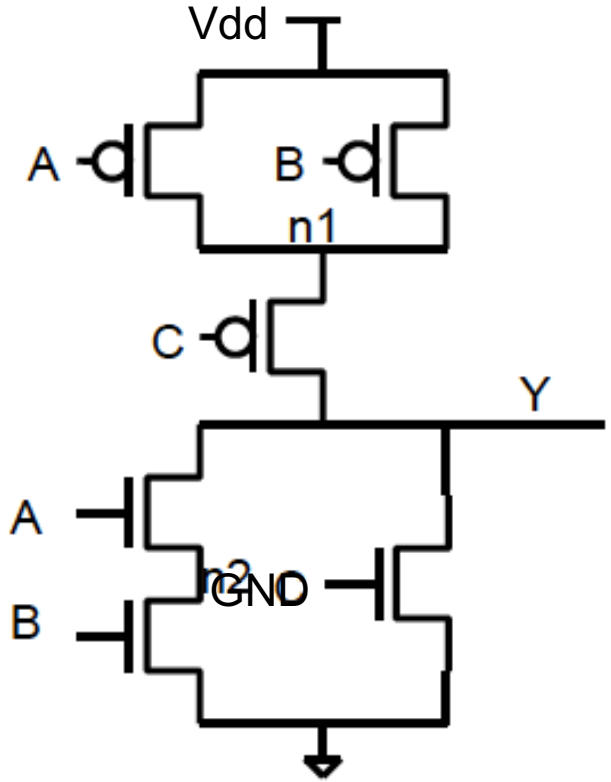


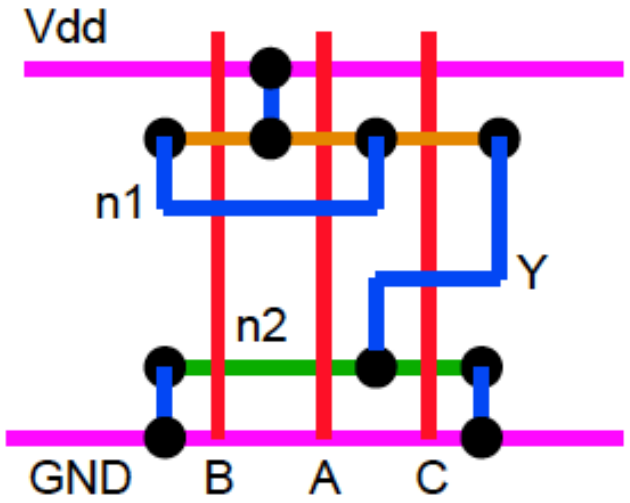


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Example - Step 5

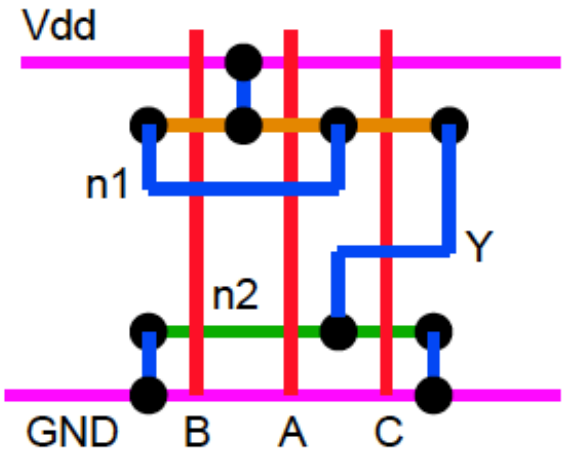
- Connect everything up

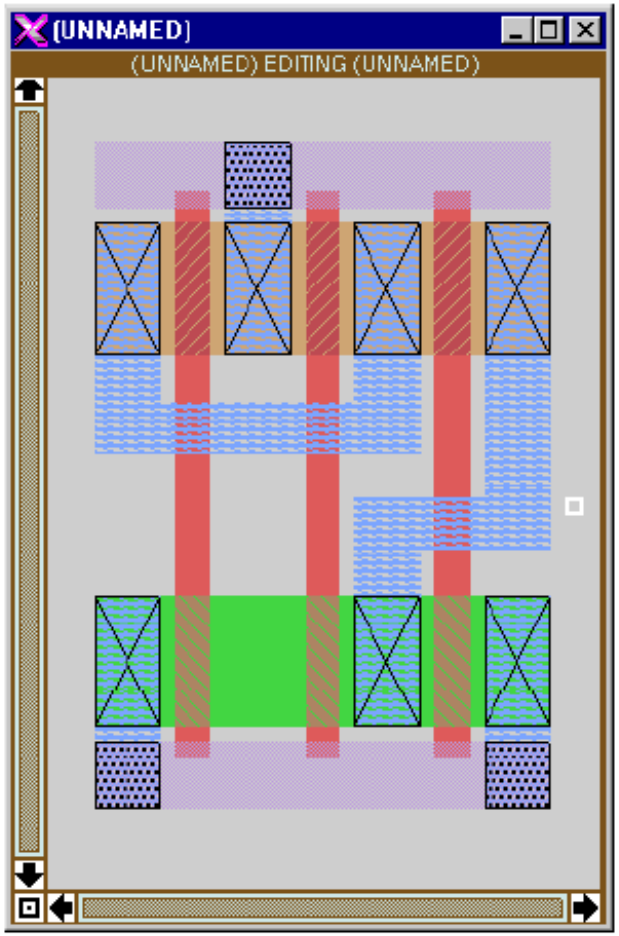




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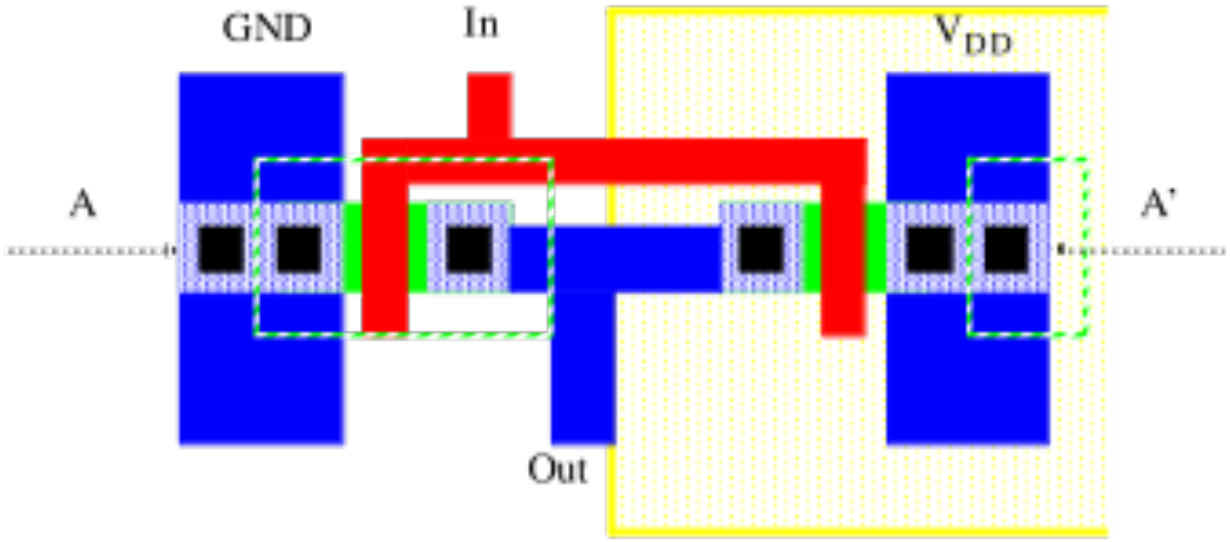
Convert to Real Layout



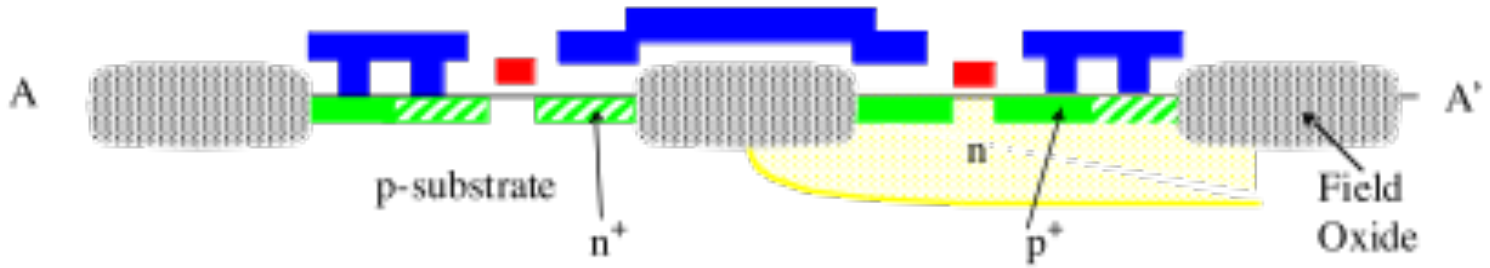


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CMOS Inverter Layout



(a) Layout



(b) Cross-Section along A-A'

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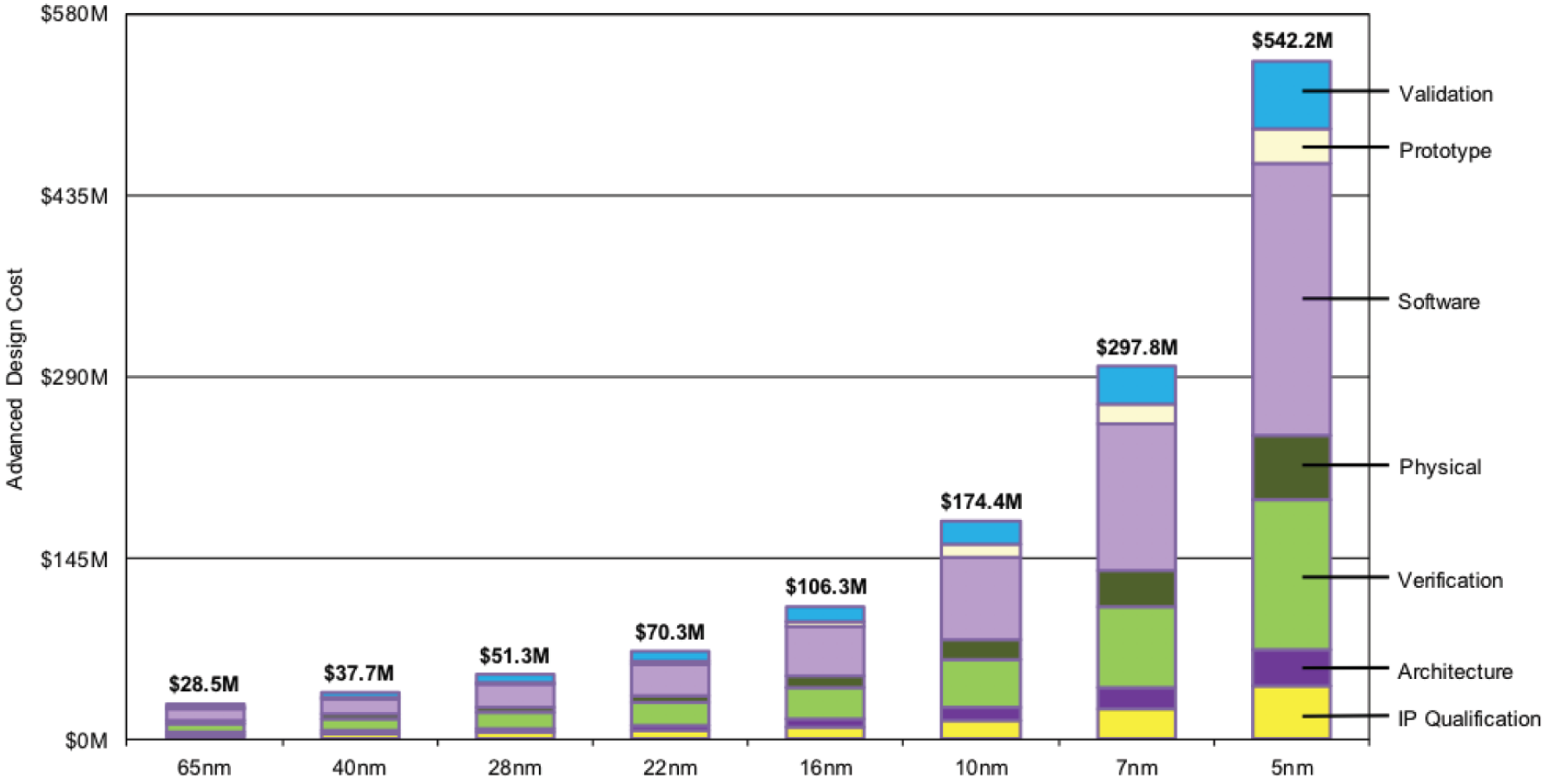
Cost of Integrated Circuits

- Non-recurring or fixed cost
  - Independent of volume
  - Examples - design time and effort, mask generation, equipment
- Recurring or variable cost
  - Dependent on the volume
  - Examples - silicon processing, packaging, test

Non-recurring cost is increasing with every new technology generation!

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Cost Of Developing New Products



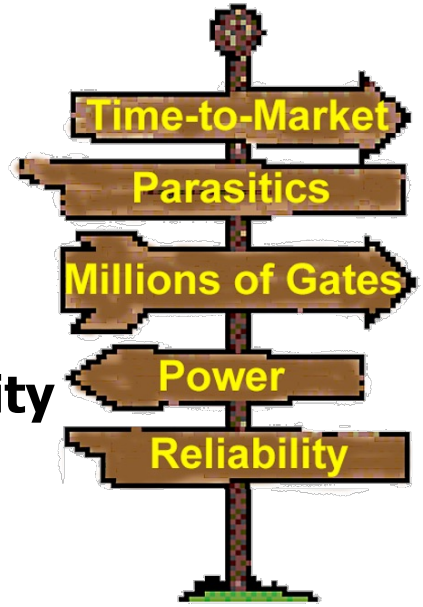
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Challenges in Digital Design

∝ DSM

"Microscopic Problems"

- Ultra-high speed design
- Interconnect
- Noise, Crosstalk
- Reliability, Manufacturability
- Power Dissipation
- Clock distribution



Everything Looks a Little Different

?

∝ 1/DSM

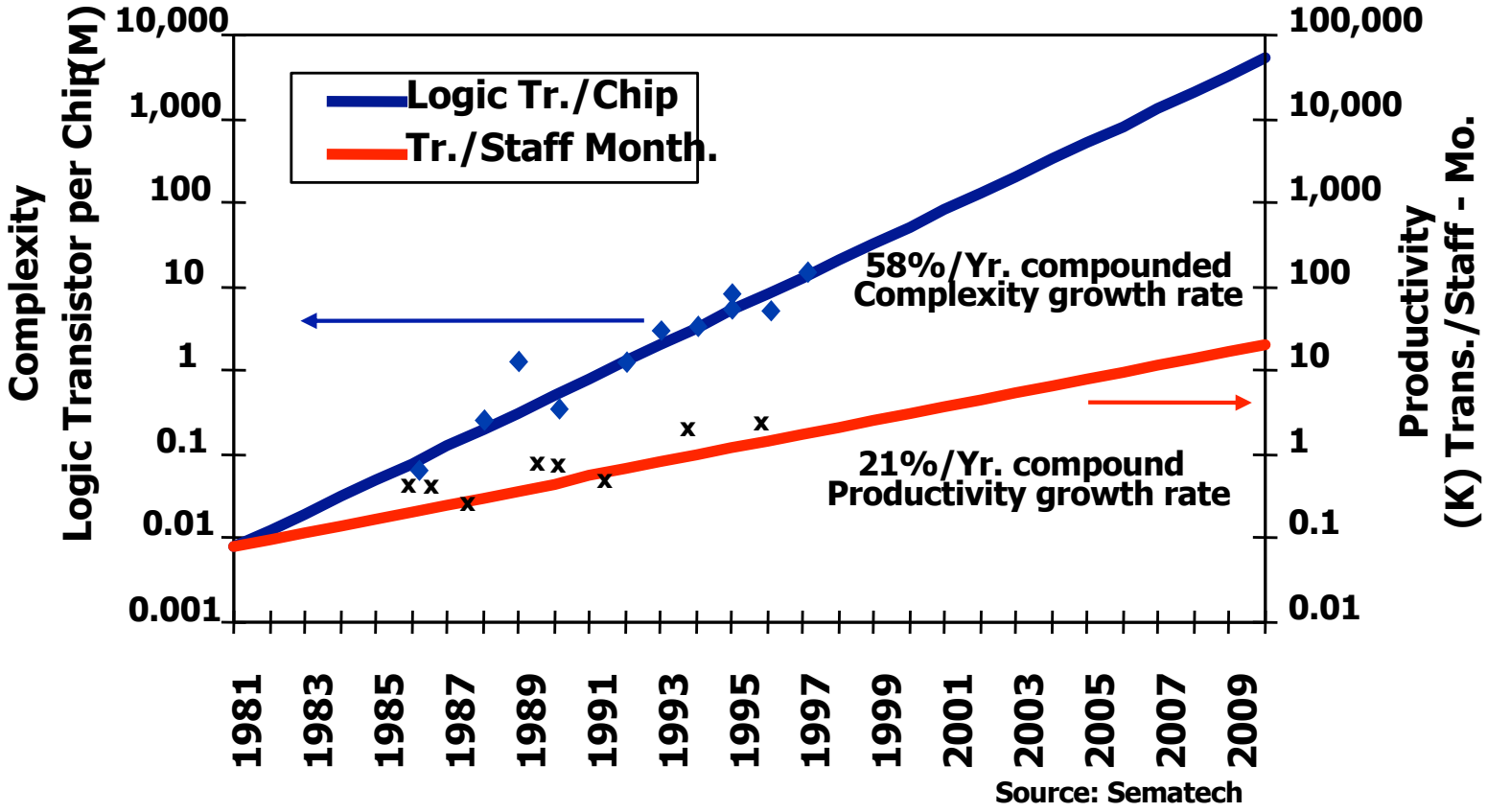
"Macroscopic Issues"

- Time-to-Market
- Billions of Gates
- High-Level Abstractions
- Reuse & IP: Portability
- Predictability
- etc.

…and There's a Lot of Them!

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Productivity Trends



Complexity outpaces design productivity

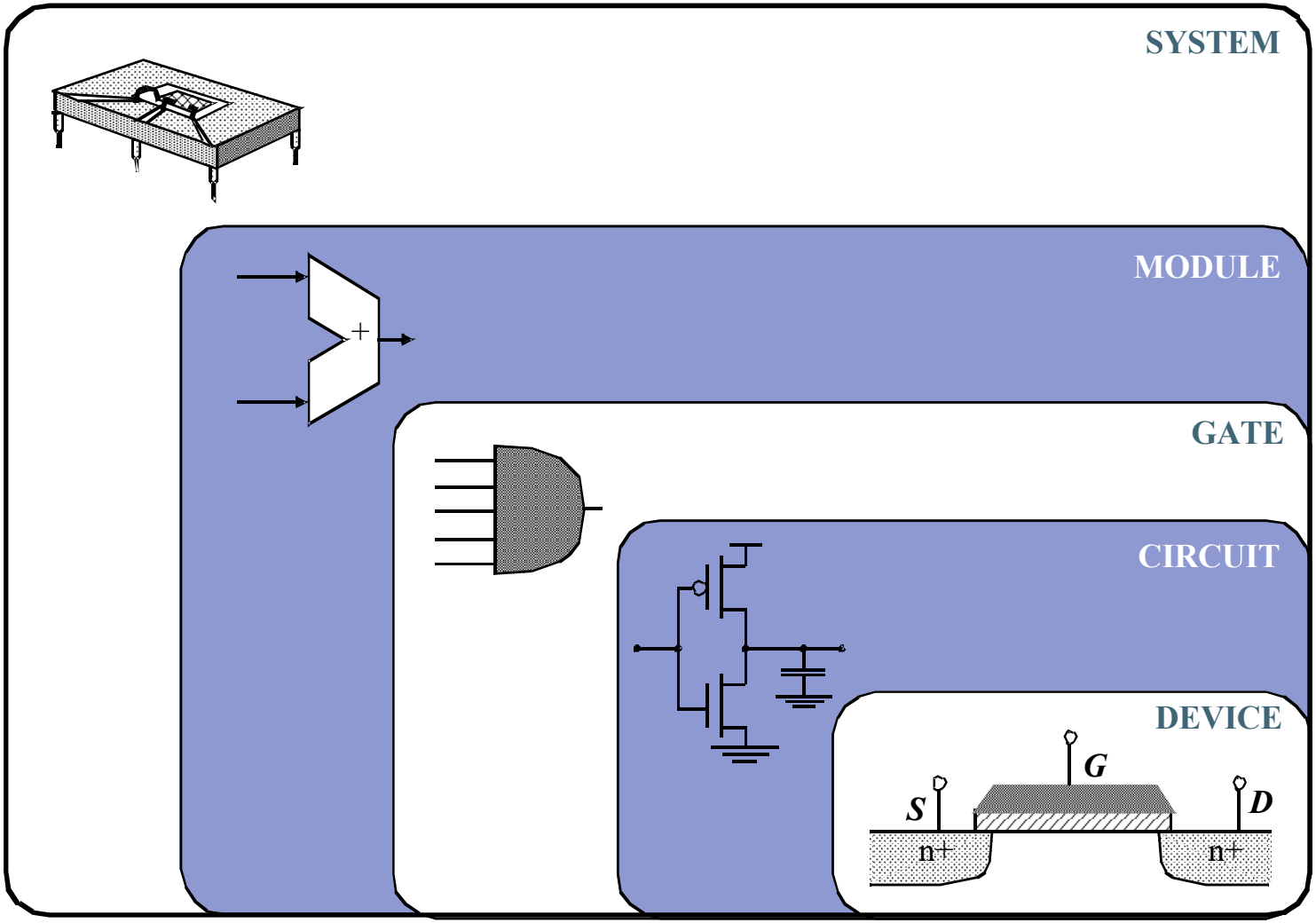
Courtesy, ITRS Roadmap

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Technology Scaling - Good or Bad?

- Rule of thumb
  - Technology shrinks by 0.7x/generation
  - With every generation we can integrate 2x more functions per chip; chip cost does not increase significantly
  - Cost of a function decreases by 2x
- But …
  - How to design chips with more and more functions?
  - Design engineering population does not double every two years…
- Hence, a need for more efficient design methods
  - Exploit different levels of abstraction

Design Abstraction Levels



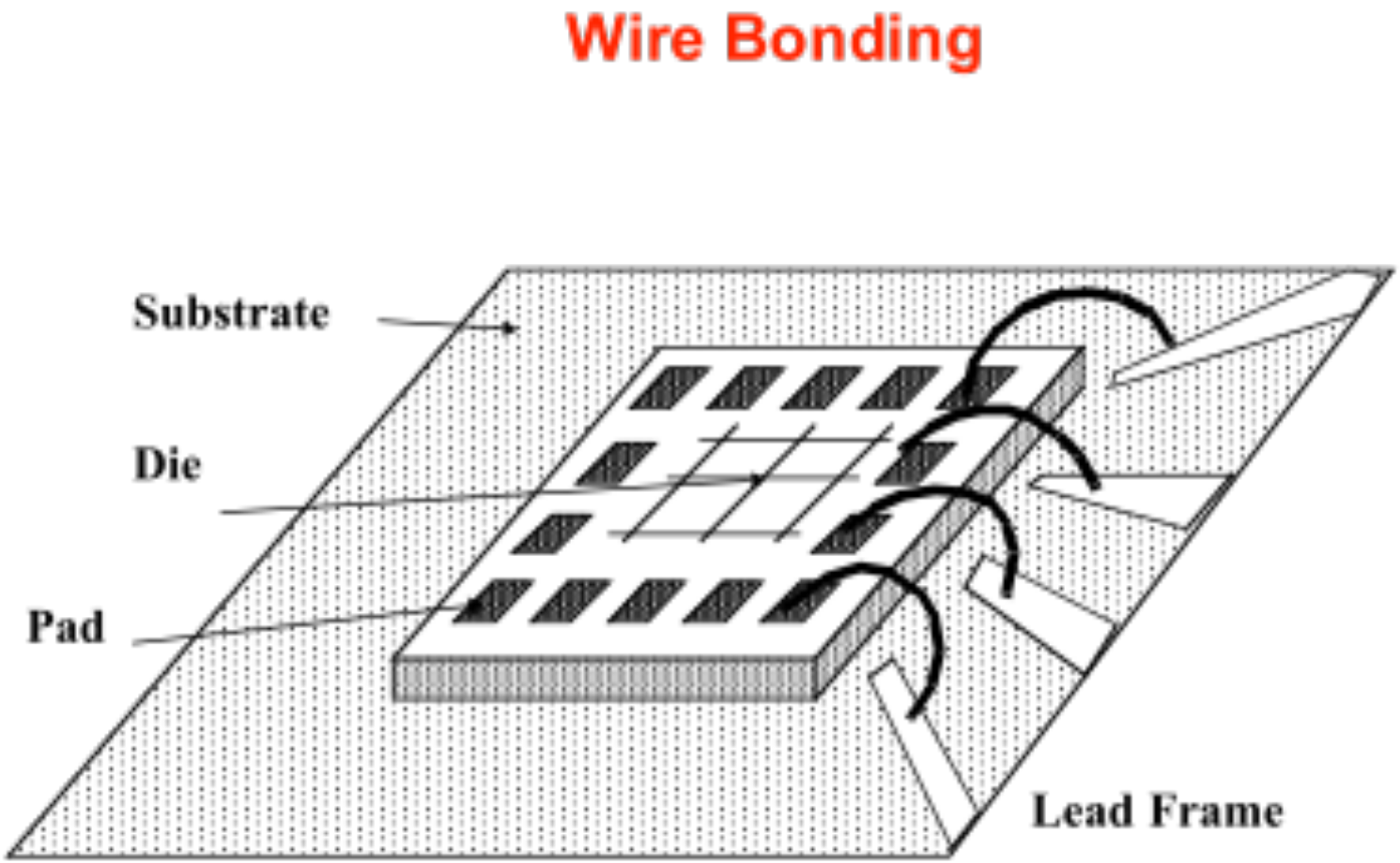
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Packaging Requirements

- Electrical: Low parasitics
- Mechanical: Reliable and robust
- Thermal: Efficient heat removal
- Economical: Cheap

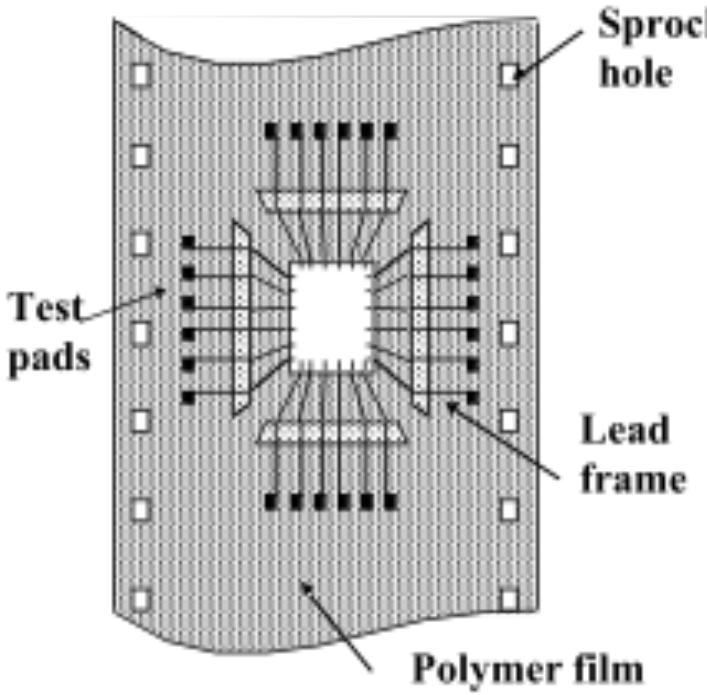
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Bonding Techniques

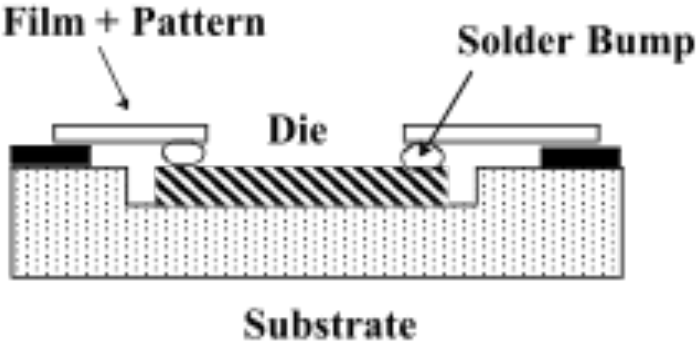


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Tape-Automated Bonding (TAB)



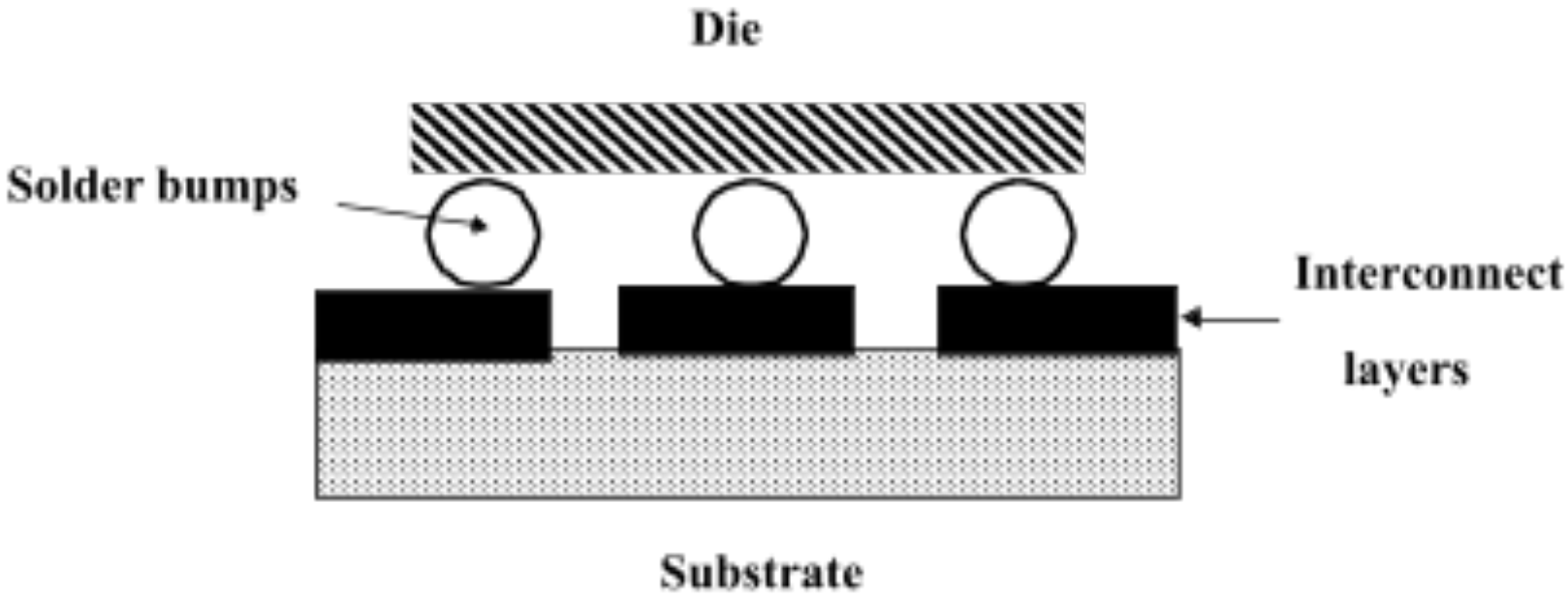
(a) Polymer Tape with imprinted wiring pattern.



(b) Die attachment using solder bumps.

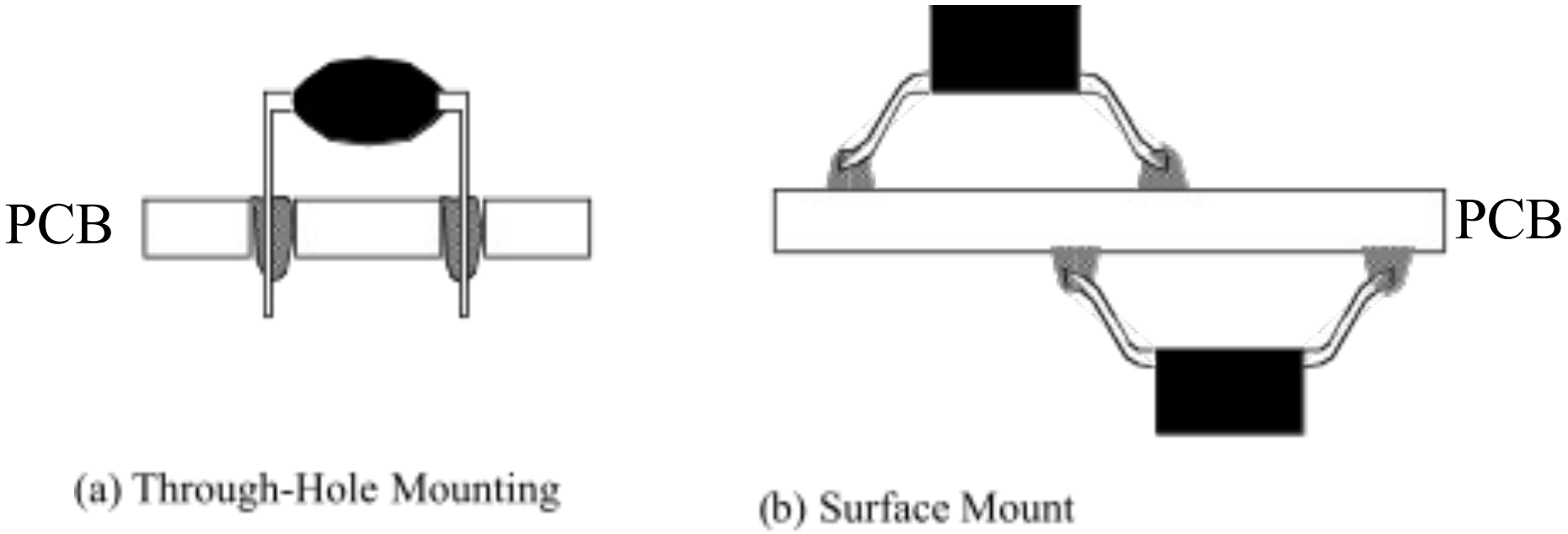
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Flip-Chip Bonding



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Package-to-Board Interconnect



(a) Through-Hole Mounting

(b) Surface Mount

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